

IIT Jodhpur IMAT MBA (Technology) - Question Bank

Time: 2 Hours | Maximum Marks: 204

Test Pattern

- **VARC:** 24 questions | 40 min
- **DILR:** 22 questions | 40 min
- **QA:** 22 questions | 40 min
- **Total:** 68 questions across 3 sections | Duration 2 Hours | Maximum Marks 204

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Each section is allotted 40 minutes; the total duration is 2 Hours.
- Attempt the sections in the given order and manage your time across all sections.

This question bank holds 136 questions -- MBA 136 (48 VARC/44 DILR/44 QA). Each test is assembled from it following the pattern above. Correct option shown in green.

Section: VARC (40 min)

Instructions (MBA1-VARC-RC1): study the shared passage below and answer its 5 question(s).

Economists have long treated their instruments as mirrors: a price index reflects inflation, an unemployment rate reflects idleness, a credit score reflects the risk that a borrower will default. The metaphor is convenient and, within limits, harmless. It becomes misleading the moment a measure acquires an audience. Once a credit score governs access to a mortgage, borrowers begin to manage the score rather than the prudence it was built to detect; lenders, in turn, price loans off the managed score. What began as a description of behaviour becomes one of the pressures that shape it. The measure is not merely inaccurate; it is productive.

It is tempting to read this as an indictment of quantification, and some critics do. The inference is too quick. Nothing in the argument shows that unmeasured judgment is more faithful to the world; loan officers working on intuition were biased in ways no one audited, and their errors, being invisible, were also uncorrectable. What the argument establishes is narrower and more awkward: the accuracy of a social measure is not a fixed property of the measure but a function of how consequential it has been allowed to become. A statistic used only by researchers may describe reasonably well what it claims to describe. The same statistic, once written into a contract or a regulation, begins to erode its own validity, and does so at a rate that has little to do with the care taken in its construction.

This has an uncomfortable corollary for institutional design. If validity decays with use, then the standard remedy, refining the metric, addresses the wrong variable. Better specification may buy time; it cannot arrest the process, because the process is driven not by measurement error but by the incentives that consequence creates. The alternatives are unattractive in different ways: rotate metrics before they are gamed, and lose comparability across periods; keep several imperfect indicators in play, and abandon the crisp accountability a single number provides; or make the metric one input among several to a judgment that remains discretionary, and therefore hard to challenge. None of these is a solution. Each is a choice about which kind of failure a system prefers to live with.

Q1. [MBA1-VARC-RC1 · mcsc · diff 0.5]

The passage is primarily concerned with establishing which of the following?

- A) Credit scoring shows how a poorly constructed statistic can end up distorting the very market that comes to depend on it.
- B) Because quantitative indicators are routinely gamed, consequential decisions are better left to experienced human judgment.
- C) Social measures fail chiefly because those who design them pay too little attention to how the underlying quantity is defined.
- D) A measure's validity falls once it is given consequences, so refining its design cannot fix the problem. ✓

Answer: D) A measure's validity falls once it is given consequences, so refining its design cannot fix the problem.

Solution: The passage argues that accuracy is a function of how consequential a measure has been allowed to become, and that refining the metric addresses the wrong variable. The appeal to human judgment is blocked by the paragraph on unaudited loan officers. Credit scoring is one illustration, not the thesis, and the passage explicitly says decay has little to do with care in construction.

Q2. [MBA1-VARC-RC1 · mcsc · diff 0.5]

The reference to loan officers working on intuition serves mainly to:

- A) suggest that decisions taken without written records can never be usefully audited
- B) establish that discretionary human judgment was historically more faithful to the world
- C) illustrate the process by which lenders learn to price loans off a managed score
- D) block the conclusion that dispensing with measurement would improve accuracy ✓

Answer: D) block the conclusion that dispensing with measurement would improve accuracy

Solution: The sentence follows 'The inference is too quick' and exists to deny the anti-quantification reading: unmeasured judgment was biased and its errors were uncorrectable. Saying intuition was more faithful reverses the point; the pricing of managed scores belongs to the earlier example; and 'never be usefully audited' overstates a remark about invisibility.

Q3. [MBA1-VARC-RC1 · mcsc · diff 1]

The author would agree with all of the following EXCEPT:

- A) A well specified metric retains its validity indefinitely if it is applied consistently. ✓
- B) A statistic may describe its object reasonably well while its use stays confined to research.
- C) Errors produced by unaudited discretion are harder to correct than errors of formal measurement.
- D) Relying on several imperfect indicators at once costs a system some of its clarity of accountability.

Answer: A) A well specified metric retains its validity indefinitely if it is applied consistently.

Solution: The passage says better specification may buy time but cannot arrest the decay, so indefinite validity from good specification is exactly what is denied. The other three restate claims made about research-only statistics, invisible and therefore uncorrectable intuition, and the loss of crisp accountability when several indicators are kept in play.

Q4. [MBA1-VARC-RC1 · mcsc · diff 1]

It can be inferred that the author regards the erosion of a measure's validity as:

- A) incentive-driven, and hence beyond the reach of purely technical refinement ✓
- B) a consequence of borrowers and lenders misunderstanding what a score measures
- C) an unavoidable feature of every attempt to describe social behaviour in numbers
- D) slower in those cases where the measure has been constructed with greater care

Answer: A) incentive-driven, and hence beyond the reach of purely technical refinement

Solution: The third paragraph attributes the process to the incentives that consequence creates rather than to measurement error, which is why refinement addresses the wrong variable. Erosion is tied to consequential use, not to all quantification, so the universal claim overreaches; the passage denies the link with care of construction; and misunderstanding is never offered as the mechanism.

Q5. [MBA1-VARC-RC1 · mcsc · diff 1]

Which of the following, if true, would most weaken the author's corollary for institutional design?

- A) Indicators published but never written into contracts decay at the same rate as those that are. ✓
- B) Regulators in practice revise their key indicators as soon as any clear evidence of widespread gaming reaches them.
- C) Firms devote considerable resources to learning exactly how scoring models weight the inputs they receive.
- D) Certain metrics embedded in regulation have gone on predicting the outcomes they were built to target for several decades.

Answer: A) Indicators published but never written into contracts decay at the same rate as those that are.

Solution: The corollary depends on consequence being the driver of decay; equal decay in indicators with no consequences attached removes that causal link and leaves refinement no worse a remedy than anything else. Decades of durability is the closest trap, but the author already concedes that specification can buy time. Revision matches the rotation option offered, and firms studying model weights is the gaming the author describes.

Instructions (MBA1-VARC-RC2): study the shared passage below and answer its 4 question(s).

Reformers who export institutions tend to work from a plausible premise: if a commercial code, a bankruptcy statute or a securities regulator produced prosperity in one place, the same text should help elsewhere. The record is not encouraging. Codes copied faithfully have been ignored, applied selectively, or absorbed into arrangements their drafters never imagined. The usual explanation is cultural: rules that suit one society's habits will not suit another's. That explanation is loose enough to accommodate any outcome, which is a reason to distrust it rather than a reason to accept it.

A tighter account starts from what a rule is for. A rule does useful work only when parties can predict how others, including officials, will respond to its breach. That prediction rests on a stock of shared expectation built up through repeated enforcement, and such a stock cannot be imported, because it is not contained in the text. A transplanted rule therefore arrives without the very thing that made it effective at home. Whether it acquires one depends less on the fit between the rule and local values than on whether somebody with something to lose finds it worth invoking, repeatedly, before officials who face some cost for ignoring it.

This reframing has consequences. It predicts that transplants will take hold in narrow commercial domains, where a few well resourced parties litigate often, before they take hold in areas of diffuse benefit, where no single stake is large enough to sustain the litigation that generates expectation. It also predicts that the most durable transplants may be the least faithful: rules bent to serve interests that already existed get enforced, while pristine copies gather dust. Reformers who measure success by fidelity to the model will therefore misread their own results, treating as failure the drift that is the mechanism of adoption.

None of this restores the reformers' optimism; it relocates the difficulty. The obstacle is not that societies are incompatible with foreign rules, but that the asset which makes rules work is produced slowly, locally, and only as a by-product of use. Where no one uses a rule, no argument about its merits will make it operative.

Q6. [MBA1-VARC-RC2 · mcsc · diff 0.5]

Which of the following comes closest to the claim the passage is defending?

- A) Reform programmes that export commercial codes to other jurisdictions should be discontinued as a matter of policy.
- B) Bankruptcy statutes and securities regulation are the two bodies of law least suited to transfer across borders.
- C) Transplants stall less from cultural mismatch than from missing shared expectation, which only local use builds. ✓
- D) Legal transplants take root only in those societies that already share the values held by the exporting country.

Answer: C) Transplants stall less from cultural mismatch than from missing shared expectation, which only local use builds.

Solution: The second and fourth paragraphs make the stock of shared expectation, which cannot travel in a text, the binding constraint, and expressly downgrade fit with local values. Requiring shared values reverses that; abandoning reform is never recommended, since the author only relocates the difficulty; and the named statutes are examples rather than a ranking.

Q7. [MBA1-VARC-RC2 · mcsc · diff 0.5]

The author's principal objection to the cultural explanation is that it:

- A) draws its evidence almost entirely from transplants that are already known to have failed
- B) confuses the substance of a rule with the circumstances under which it was drafted
- C) can be reconciled with any observed result, and so accounts for nothing in particular ✓
- D) understates the extent to which commercial habits differ from one society to another

Answer: C) can be reconciled with any observed result, and so accounts for nothing in particular

Solution: The passage calls the cultural account loose enough to accommodate any outcome and treats that looseness as grounds for distrust, which is the standard charge of unfalsifiability. Nothing is said about selection of cases, the author does not claim habits differ more than culturalists allow, and no confusion of substance with drafting is alleged.

Q8. [MBA1-VARC-RC2 · mcsc · diff 1]

On the author's account, a transplanted consumer protection rule would be expected to become operative more slowly than a transplanted secured-lending rule mainly because:

- A) consumer protection sits in sharper conflict with the commercial customs prevailing locally
- B) consumers as a group tend to be much less aware of the legal remedies formally available to them
- C) the benefit from enforcing it is spread too thinly for repeated litigation to be worthwhile ✓
- D) rules governing secured lending are drafted more simply and are easier for officials to apply

Answer: C) the benefit from enforcing it is spread too thinly for repeated litigation to be worthwhile

Solution: The mechanism given is that expectation is generated by frequent litigation, which requires a party whose stake is large enough to justify it; diffuse benefit therefore starves the process. Conflict with custom is the cultural story the author sets aside, drafting simplicity is never invoked, and low awareness is a plausible sounding substitute for the stake-size argument the passage actually makes.

Q9. [MBA1-VARC-RC2 · mcsc · diff 1]

The author would agree with all of the following EXCEPT:

- A) Departures from the imported text can form part of the very process by which a rule is adopted.
- B) How a rule works depends on what parties predict officials will do when the rule is breached.
- C) Where local values are hostile to an imported rule, enforcement cannot make it effective. ✓
- D) Transplants settle sooner in those fields where a small number of parties go to court frequently.

Answer: C) Where local values are hostile to an imported rule, enforcement cannot make it effective.

Solution: The passage makes repeated invocation before officials who bear a cost for inaction the decisive factor and treats fit with local values as secondary, so a veto by hostile values contradicts it. The other three restate the prediction-of-response account, the claim that the least faithful transplants may be the most durable, and the prediction about narrow commercial domains.

Q10. [MBA1-VARC-ODD-1 · mcsc · diff 0.5]

Five sentences are given below, labelled 1 to 5. Four of them can be arranged to form a coherent paragraph. Identify the sentence that does not belong.

- (1) Every map begins with an act of omission, since a representation that kept all the detail of a place would be as unusable as the place itself.
- (2) The cartographer must therefore decide in advance which features a reader will need and which can be silently discarded.
- (3) Recent surveys report that most commuters now consult digital route planners rather than printed atlases.
- (4) Those early decisions harden over time into conventions, and later users mistake the conventions for properties of the world.
- (5) What looks like a neutral record of terrain is thus the residue of a long series of judgements about relevance.

Which sentence is the odd one out?

- A) 1
- B) 2
- C) 3 ✓
- D) 4
- E) 5

Answer: C) 3

Solution: Sentences 1, 2, 4 and 5 build one argument: maps must omit, so a cartographer selects, the selections become conventions, and the finished map is therefore a record of judgements about relevance. Sentence 3 makes an empirical claim about which media commuters use today, which contributes nothing to the argument about selection and is the misfit. Sentence 1 is the natural opener, 2 follows from it through therefore, and 5 is the closing generalisation, so none of those can be removed without breaking the chain.

Q11. [MBA1-VARC-ODD-2 · mcsc · diff 0.5]

Five sentences are given below, labelled 1 to 5. Four of them can be arranged to form a coherent paragraph. Identify the sentence that does not belong.

- (1) Experts seldom deliberate over the moves that most impress onlookers; they recognise a familiar pattern and act before reasons are available to them.
- (2) Such fluency is bought by long exposure to situations that return prompt and unambiguous feedback.
- (3) Where the feedback is delayed or noisy, however, long experience yields confidence without yielding accuracy.
- (4) Novices, by contrast, report that they enjoy the earliest stages of learning a craft far more than the later ones.
- (5) Intuition therefore deserves trust only after one has asked what the environment has actually been teaching.

Which sentence is the odd one out?

- A) 3
- B) 4 ✓
- C) 5
- D) 1
- E) 2

Answer: B) 4

Solution: Sentences 1, 2, 3 and 5 form a single line of reasoning about when expert intuition can be trusted: it looks effortless, it depends on good feedback, it fails where feedback is poor, and the practical rule follows. Sentence 4 reports how much novices enjoy learning, a claim about affect rather than about the reliability of intuition, so it is the outsider. Sentence 3 cannot be the answer because however marks exactly the contrast that sentence 5 then resolves.

Q12. [MBA1-VARC-ODD-3 · mcsc · diff 1]

Five sentences are given below, labelled 1 to 5. Four of them can be arranged to form a coherent paragraph. Identify the sentence that does not belong.

- (1) A quantity chosen to stand for a goal starts to be pursued for its own sake as soon as rewards are attached to it.
- (2) Staff learn quickly which parts of their work are counted, and effort drains away from everything that is not.
- (3) The published figures then improve while the purpose they were meant to serve quietly deteriorates.
- (4) Organisations often adopt measurement schemes less to inform their own decisions than to demonstrate propriety to outside audiences.
- (5) Auditors, reading only the figures, record success and recommend that the scheme be extended to further departments.

Which sentence is the odd one out?

- A) 4 ✓
- B) 5
- C) 2
- D) 3
- E) 1

Answer: A) 4

Solution: Sentences 1, 2, 3 and 5 trace one causal sequence: attaching rewards to an indicator redirects effort, the indicator rises as the underlying purpose decays, and auditors who see only the indicator extend the scheme. Sentence 4 stays on the same subject but shifts the level of analysis, explaining why organisations take up measurement at all rather than what measurement does to behaviour, and it plays no part in the causal chain. Sentence 5 belongs because it supplies the consequence of the improved figures named in sentence 3.

Q13. [MBA1-VARC-ODD-4 · mcsc · diff 1]

Five sentences are given below, labelled 1 to 5. Four of them can be arranged to form a coherent paragraph. Identify the sentence that does not belong.

- (1) No translator can carry every feature of a sentence across, so the work consists largely in deciding what to give up.
- (2) Readers tend to praise the translations that read most smoothly and to distrust those that preserve the strangeness of the original.
- (3) A pun that depends on sound may be replaced by one that depends on sense, on the ground that the joke matters more than its mechanism.
- (4) Rhythm surrendered in one line is sometimes repaid in another, the debt being settled across the page rather than within it.
- (5) Every such compensation is an argument about which layer of the original counts as the text itself.

Which sentence is the odd one out?

- A) 1
- B) 4
- C) 5
- D) 2 ✓
- E) 3

Answer: D) 2

Solution: Sentences 1, 3, 4 and 5 describe the translator's own practice: loss is unavoidable, so features are traded and debts are repaid elsewhere, and each trade amounts to a claim about what the text essentially is. Sentence 2 describes how readers receive finished translations, a claim about reception rather than about the decisions being made, and the paragraph never uses it. Sentence 4 is not the misfit because the repayment it describes is one instance of the compensation that sentence 5 generalises.

Q14. [MBA1-VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) That demand is what makes boredom productive, since it pushes the mind to supply what the surroundings withhold.
- (2) Boredom is usually described as an absence, the gap left when nothing outside us claims our attention.
- (3) But the state has a content of its own, a restless demand that the world become more interesting than it is.
- (4) Children left with nothing to do therefore stay idle only briefly, building a game out of whatever lies nearest.

Which of the following is the correct order of the sentences?

- A) 2341
- B) 2314 ✓
- C) 3241
- D) 2134

Answer: B) 2314

Solution: Sentence 2 gives the standard description and must open. Sentence 3 begins with But and corrects that description by introducing the restless demand. Sentence 1 picks that demand up with That demand, so it can only follow. Sentence 4 closes with therefore and an illustration. The order beginning 2134 fails because That demand has no antecedent yet, 3241 puts the correction before what it corrects, and 2341 leaves sentence 1 stranded after the example it explains.

Q15. [MBA1-VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) Once the date is fixed, the remaining hours acquire a rank, and the work is fitted to those that are left.
- (2) This is why arbitrary dates can serve as well as reasoned ones, provided they are actually believed.
- (3) A deadline works less by adding motivation than by removing alternatives.
- (4) Until a date is fixed, every hour is a candidate for the task and no hour is obliged to be.

Which of the following is the correct order of the sentences?

- A) 3421
- B) 3142
- C) 3412 ✓
- D) 4312

Answer: C) 3412

Solution: Sentence 3 states the thesis about removing alternatives and opens. Sentences 4 and 1 form the matched pair Until a date is fixed and Once the date is fixed, in that order, since the second echoes the first. Sentence 2 draws the inference with This is why and must end. The order 3142 breaks the Until and Once pair, 4312 buries the thesis, and 3421 places the inference before the mechanism it depends on.

Q16. [MBA1-VARC-JUMBLE-3 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) The only reliable lever, then, is conduct that gives those custodians something else to carry.
- (2) Its custodians are third parties, who pass it on for reasons of their own and revise it without notice.
- (3) Reputation is information about us that we do not own and cannot directly edit.
- (4) This dependence explains why attempts at correction so often deepen the damage, for a correction is itself testimony and it travels.

Which of the following is the correct order of the sentences?

- A) 3421
- B) 2431
- C) 3241 ✓
- D) 2341

Answer: C) 3241

Solution: Only sentence 3 introduces reputation as a full noun, so the pronoun Its in sentence 2 must come after it. Sentence 4 opens with This dependence, which refers to the reliance on third parties just established, and sentence 1 closes with then and with those custodians. The order 2341 and 2431 both strand the pronoun Its ahead of its antecedent, and 3421 makes This dependence point at nothing but the definition.

Q17. [MBA1-VARC-JUMBLE-4 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) A first reading of anything difficult is mostly navigation, since we are still learning where the parts are.
- (2) Readers who skip it are not saving time; they are trading comprehension for the feeling of having finished.
- (3) For this reason a second reading is not a repetition but a different task, undertaken with the map the first reading drew.
- (4) Only once the shape is known can attention be spent on the joints, which is where arguments usually fail.

Which of the following is the correct order of the sentences?

- A) 4132
- B) 1342
- C) 1432 ✓
- D) 1423

Answer: C) 1432

Solution: Sentence 1 must open, because Only once the shape is known in sentence 4 presupposes the navigating first reading. Sentence 3 then draws the conclusion with For this reason, and the pronoun it in sentence 2 refers to the second reading named there, so sentence 2 comes last. The order 1342 puts the conclusion before the reason, 4132 opens with a clause that has nothing to refer back to, and 1423 leaves it without its antecedent.

Q18. [MBA1-VARC-JUMBLE-5 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, are not in order. Arrange them into a logical paragraph.

- (1) The reason travels less well than the exception, since the reason was particular and the exception is not.
- (2) What began as mercy is then read as entitlement, and the rule survives only as a memory.
- (3) Administrators who grasp this record their reasons loudly, hoping the record will do the limiting work the rule no longer can.
- (4) Every exception granted for a good reason becomes a fact that later applicants are free to cite.

Which of the following is the correct order of the sentences?

- A) 4213
- B) 1423
- C) 4132
- D) 4123 ✓

Answer: D) 4123

Solution: Sentence 4 must open, because the definite phrase The reason in sentence 1 refers back to the good reason mentioned there. Sentence 2 follows with then, marking the end point of the drift, and sentence 3 responds to the whole process with this. The order 1423 begins with a reference that has no antecedent, 4213 reverses the drift and its explanation, and 4132 makes this point at an explanation rather than at the completed process.

Q19. [MBA1-VARC-SUMMARY-1 · mcsc · diff 0.5]

Infrastructure is noticed chiefly when it fails. A water main, a switching station, a clearing house: each is built to disappear into the routines it supports, and its success is measured by how little thought it demands. The consequence is political rather than technical. Systems nobody thinks about attract neither scrutiny nor investment, and their upkeep competes badly with projects that can be photographed. Failure then arrives as a surprise, and the surprise is used to argue that nobody could have foreseen it.

Which option best summarises the paragraph?

- A) Public money should be redirected from photogenic new construction towards the unglamorous upkeep of water mains, switching stations and clearing houses.
- B) Infrastructure fails because engineers deliberately design systems to disappear, and systems that are invisible are inherently more fragile than visible ones.
- C) Since infrastructure failures are always surprises, the political value of such systems lies in the attention that their breakdowns finally bring to them.
- D) Designed to go unnoticed, working infrastructure draws little scrutiny or funding, so its breakdowns look unforeseeable when they were merely unattended. ✓

Answer: D) Designed to go unnoticed, working infrastructure draws little scrutiny or funding, so its breakdowns look unforeseeable when they were merely unattended.

Solution: The paragraph links invisibility to neglect and neglect to apparently unforeseeable failure, which is exactly what the summary about unattended breakdowns captures. The claim that invisible systems are inherently more fragile substitutes a technical mechanism the paragraph explicitly rejects. The call to redirect public money is a recommendation the paragraph never makes. Treating breakdowns as valuable attention inverts the point that the surprise is used as an excuse.

Q20. [MBA1-VARC-SUMMARY-2 · mcsc · diff 0.5]

Averaging many independent guesses often beats the best single guess, because errors pointing in opposite directions cancel. The condition doing the work is independence. Once the guessers can hear one another, their errors begin to align, and the average inherits a bias instead of shedding one. Committees that deliberate before voting therefore give up much of the advantage voting was supposed to secure, and the confidence produced by agreement rises just as the accuracy it is taken to signal falls.

Which option best summarises the paragraph?

- A) Confidence is a poor guide to accuracy, since agreement inside a group can show only that its members have all been exposed to the same limited information.
- B) Groups outperform their ablest members because pooling opinions cancels error, which is why committees ought always to vote instead of trusting one expert.
- C) Aggregation improves accuracy only while guesses stay independent; discussion correlates errors, so deliberating groups gain confidence and lose accuracy. ✓
- D) Deliberation harms committees chiefly because the members who happen to speak first impose their own private biases on all of those who speak later.

Answer: C) Aggregation improves accuracy only while guesses stay independent; discussion correlates errors, so deliberating groups gain confidence and lose accuracy.

Solution: The paragraph makes independence the load-bearing condition and ends on the divergence between rising confidence and falling accuracy, which the summary naming both captures. The version recommending that committees always vote keeps the cancelling of errors but drops the condition that makes it work. The claim about who speaks first invents a mechanism. The remark about confidence alone reports one consequence and omits independence entirely.

Q21. [MBA1-VARC-SUMMARY-3 · mcsc · diff 0.5]

Options are usually treated as free, on the view that holding one costs nothing until it is exercised. In practice every open possibility levies a quiet charge: some attention reserved, some commitment withheld, some relationship kept provisional. Because the charge is never invoiced, options accumulate well past the point where their combined upkeep exceeds anything they could deliver. The person who cannot choose is not made richer by the alternatives; the alternatives are consuming the very capacity that choosing would have released.

Which option best summarises the paragraph?

- A) Open possibilities exact hidden upkeep in attention and commitment, so options collected beyond a point cost more than they could ever deliver. ✓
- B) Because an option costs nothing at all until it is exercised, a rational person should hold as many as possible and delay commitment for as long as possible.
- C) Indecisive people are poorer than decisive ones, because real wealth consists in commitments actually made rather than in alternatives that remain available.
- D) The chief trouble with unexercised options is that their true price is never invoiced, which makes any accounting of their value unreliable in practice.

Answer: A) Open possibilities exact hidden upkeep in attention and commitment, so options collected beyond a point cost more than they could ever deliver.

Solution: The paragraph identifies a hidden cost of holding options and locates a threshold past which the cost outweighs the benefit, which the summary about upkeep exceeding delivery states. The advice to hold as many options as possible repeats the view the paragraph is arguing against. The claim about indecisive people being poorer generalises into a thesis about wealth that the text does not assert. The uninvoiced price is only the reason the cost goes unnoticed, not the main point.

Q22. [MBA1-VARC-SUMMARY-4 · mcsc · diff 1]

Forecasts about human affairs are not merely difficult; they are ingredients. A published prediction of shortage produces hoarding, and a published prediction of stability invites the risk-taking that ends it. This does not make forecasting useless, but it changes what accuracy could mean, since a forecast can be judged only against the world it helped to make, and a forecast that is widely ignored may be the only one ever fairly tested. Practitioners who forget this treat their failures as errors of measurement and answer them by refining instruments that were never the problem.

Which option best summarises the paragraph?

- A) Predictions about human affairs are self-fulfilling, so forecasters should publish nothing until their instruments are accurate enough to withstand contact with the public.
- B) Predictions about human affairs alter the conditions they describe, so accuracy can be judged only against an already altered world, and better instruments cannot help. ✓
- C) Social forecasts change the behaviour they describe, which is why only those forecasts that nobody happens to notice can ever be tested fairly against events.
- D) Failures of social forecasting come from measurement error, and sharper instruments will in time let practitioners predict hoarding and risk-taking reliably enough.

Answer: B) Predictions about human affairs alter the conditions they describe, so accuracy can be judged only against an already altered world, and better instruments cannot help.

Solution: The paragraph's main claim is that because forecasts feed back into events, accuracy must be redefined and better instruments cannot rescue the enterprise, which is the summary mentioning both the altered world and the instruments. The version about forecasts nobody notices repeats a subordinate aside and leaves out the conclusion about measurement. The advice to publish nothing adds a prescription and treats every forecast as self-fulfilling, which the shortage and stability cases contradict. Blaming measurement error states the mistake the paragraph diagnoses.

Q23. [MBA1-VARC-SUMMARY-5 · mcsc · diff 1]

A standard is valuable in proportion to how many people already use it, so its merits and its market share are hard to tell apart. Once adoption passes a threshold, the standard is defended by everything built on top of it, and a technically superior rival must compensate not for a small gap in quality but for the whole accumulated weight of compatibility. Arguments about which standard is better therefore arrive too late to matter, and the winner's supporters read its survival as evidence that the argument was settled on the merits.

Which option best summarises the paragraph?

- A) Technically superior standards displace entrenched ones eventually, provided their supporters can show that the original choice was never really settled on the merits.
- B) A standard's worth grows with its user base, so entrenched standards defeat better rivals through compatibility, and survival is then misread as proof of merit. ✓
- C) Because compatibility counts for more than quality, users should simply adopt whichever standard is most widely used instead of assessing competing designs themselves.
- D) A standard's dominance owes far more to the number of people using it than to its quality, so late arguments about technical merit rarely change which one prevails.

Answer: B) A standard's worth grows with its user base, so entrenched standards defeat better rivals through compatibility, and survival is then misread as proof of merit.

Solution: The paragraph moves from network value to compatibility lock-out and ends with the mistaken inference from survival to superiority, all three of which the summary about misread survival preserves. The version about late arguments is accurate as far as it goes but stops before that inference, which is the paragraph's sting. The claim that superior standards eventually win reverses the argument. The advice to adopt the most popular standard converts a diagnosis into guidance the text does not offer.

Q24. [MBA1-VARC-SUMMARY-6 · mcsc · diff 1]

An archive looks like a residue of the past, but it is a residue of past decisions about what would be worth keeping. Records survive because someone was obliged to file them, and the obligations fell unevenly, heavily on property and litigation and hardly at all on ordinary agreement. Historians who read the surviving mass as a sample of what happened therefore reconstruct a world of disputes and transactions and miss the settled life around them. The remedy is not more documents but a habit of asking, of every silence, whether it records an absence or only an exemption from paperwork.

Which option best summarises the paragraph?

- A) Because filing obligations bore most heavily on property and litigation, surviving archives over-represent disputes and transactions at the expense of ordinary agreement.
- B) Every silence in an archive records a genuine absence, which is why the past appears to consist mainly of litigation, transactions and disputes rather than agreement.
- C) Historians ought to seek out further documents about settled everyday life, since the archives that do survive were shaped by the interests of owners and courts.
- D) Archives keep whatever filing duties happened to capture, so reading them as a fair sample distorts the past unless each silence is itself interrogated. ✓

Answer: D) Archives keep whatever filing duties happened to capture, so reading them as a fair sample distorts the past unless each silence is itself interrogated.

Solution: The paragraph diagnoses a bias created by uneven filing duties and then prescribes a way of reading silences, and only the summary combining distortion with interrogating silences carries both halves. The statement about property and litigation is true but stops at the diagnosis and omits the remedy. The suggestion to seek more documents is precisely the remedy the text rules out. The assertion that every silence records a real absence states the error the paragraph warns against.

Instructions (MBA2-VARC-RC1): study the shared passage below and answer its 5 question(s).

When we ask what a poem means, we are tempted to reach for what its author intended it to mean, as though the work were a sealed letter and criticism the act of opening it. The temptation is understandable, since intentions do govern most of what we make and say. Yet a strand of twentieth-century criticism insisted that, once a work is finished and released, the author's private aims cease to be the arbiter of its meaning. On this view the poem is a public object made of language, and language carries implications its user may neither have foreseen nor endorsed. A reader who recovers a documented intention has learned something about the poet's biography, not necessarily something decisive about the poem.

The argument is easy to caricature as a licence for arbitrary reading, but its careful proponents claimed no such thing. They distinguished the meaning of a work from the occasion of its composition, and held that evidence internal to the text, its diction, structure, and the conventions it invokes, constrains interpretation more tightly than any appeal to what the author confided in a letter. Where internal and external evidence agree, the appeal to intention is redundant; where they conflict, it is the text, not the confession, that the critic is professionally bound to honour. Intention, in short, is neither available nor desirable as a final standard of literary judgement.

Critics of this position have replied that meaning without a meaner is incoherent, and that to strip intention from interpretation is to reduce reading to the free play of association. The reply has force against the caricature but less against the original claim, which never denied that someone meant something, only that the someone's report settles the question. A more searching objection notes that the very conventions the text invokes are themselves the sediment of countless past intentions, so that intention returns, collectivised, at the level of the language itself. The debate, then, is less about whether intention matters than about whose intention, and at what remove, a competent reading is obliged to reconstruct.

Q25. [MBA2-VARC-RC1 · mcsc · diff 0.5]

The passage is primarily concerned with:

- A) defending the claim that a reader is free to assign a poem almost any meaning that the language will loosely permit
- B) tracing how twentieth-century literary criticism gradually abandoned nearly all close attention to the words on the page
- C) presenting a view that denies the author's intention fixes a poem's meaning, and weighing the replies it draws ✓
- D) arguing that an author's stated intention is wholly irrelevant to every form of literary and biographical inquiry

Answer: C) presenting a view that denies the author's intention fixes a poem's meaning, and weighing the replies it draws

Solution: The passage lays out the anti-intention position (paragraphs 1 to 2) and then the objections to it (paragraph 3), ending on what the dispute is really about. It does not claim intention is irrelevant everywhere; it grants intention is biographically informative and even returns collectivised, so the 'wholly irrelevant' option overstates. The 'any meaning' option is the caricature the passage explicitly rejects. The historical-abandonment option is a scope shift the text never makes.

Q26. [MBA2-VARC-RC1 · mcsc · diff 0.5]

The careful proponents of the anti-intention view would agree with all of the following EXCEPT:

- A) The diction and structure of a text limit its readings more tightly than a private confession by its author
- B) Where internal and external evidence point the same way, invoking the author's intention adds nothing new
- C) A finished poem has no determinate meaning until its author supplies an authoritative gloss on it ✓
- D) Recovering an author's documented intention can still teach us something about that author's biography

Answer: C) A finished poem has no determinate meaning until its author supplies an authoritative gloss on it

Solution: The proponents hold that the text, not the author's gloss, fixes meaning, so the claim that a poem lacks meaning until the author supplies an authoritative gloss inverts their position and is the EXCEPT. The other three restate points made directly: intention is biographically informative, it is redundant when the evidence agrees, and internal evidence constrains reading more than a confession.

Q27. [MBA2-VARC-RC1 · mcsc · diff 1]

It can be inferred that the 'more searching objection' differs from the simpler reply that precedes it because it:

- A) abandons the appeal to textual evidence and defends interpretation as the reader's free play of association
- B) denies outright that any individual author ever meant anything determinate by the words chosen for a poem
- C) rests on the premise that a work can carry no meaning at all unless a single mind stands directly behind it
- D) grants the priority given to the text while relocating intention to the conventions the language carries ✓

Answer: D) grants the priority given to the text while relocating intention to the conventions the language carries

Solution: The searching objection accepts that the text constrains reading yet points out that the conventions are themselves 'the sediment of countless past intentions', so intention returns collectivised rather than being reasserted for the lone author. The 'meaning without a meaner' premise belongs to the simpler reply, not this one. The other options attribute denials or a free-play stance that the objection does not make.

Q28. [MBA2-VARC-RC1 · mcsc · diff 1]

The author calls the appeal to intention 'redundant' when evidence agrees and to be overruled when evidence conflicts chiefly in order to:

- A) show that on either alignment of the evidence the author's intention does no decisive work ✓
- B) establish the separate point that authors are habitually unreliable witnesses to their own real aims
- C) suggest that external evidence about a work is nearly always misleading and is best set entirely aside
- D) concede that in a narrow range of disputes an author's own stated intention does settle the reading

Answer: A) show that on either alignment of the evidence the author's intention does no decisive work

Solution: The two-case structure is a dilemma: if internal and external evidence agree, intention is superfluous; if they conflict, the text wins anyway. Either way intention is not the deciding factor, which is the rhetorical work the sentence performs. It does not concede that intention sometimes settles the reading, nor argue authors are unreliable witnesses, nor dismiss external evidence wholesale.

Q29. [MBA2-VARC-RC1 · mcsc · diff 1]

Which of the following claims about the debate is fully supported by the passage?

- A) The passage shows that intention of every kind can be dispensed with once a text is complete
- B) The live question is which intentions, and how far mediated, a competent reading must reconstruct ✓
- C) The critics have demonstrated that the anti-intention view collapses into arbitrary reading
- D) The passage concludes that an individual author's intention proves indispensable to interpretation after all

Answer: B) The live question is which intentions, and how far mediated, a competent reading must reconstruct

Solution: The final sentence recasts the dispute as one about whose intention and at what remove a reading must reconstruct, which the correct option states exactly. The 'indispensable after all' option overreads the collectivised return of intention as a rescue of the individual author, which the text does not grant. The 'dispensed with entirely' option ignores that collectivised return. The 'collapses into arbitrary reading' option is the caricature the passage says has force only against a straw version.

Instructions (MBA2-VARC-RC2): study the shared passage below and answer its 4 question(s).

For a generation, the standard verdict on shared resources was pessimistic: where no one owns a pasture, a fishery, or an aquifer, each user has an incentive to take as much as possible before a rival does, and the resource is ruined by the sum of individually rational choices. The remedy seemed to follow with the force of a theorem. Either the state must regulate the resource from above, or it must be carved into private parcels whose owners will husband what they hold. Between the regulator and the auctioneer there appeared to be no third path.

Field research complicated the theorem. In mountain villages, coastal fisheries, and irrigation systems that had endured for centuries, users had neither nationalised nor privatised their commons, yet had not destroyed them either. Instead they had crafted their own rules: who could harvest, when, how much, and what would befall a cheat. These arrangements were neither spontaneous nor costless. They depended on the users being able to monitor one another, to communicate, to revise the rules as conditions changed, and to impose graduated penalties that punished a first offence lightly and a repeated one severely. Where such conditions were absent, the pessimistic prediction often held.

The lesson was not that the original model was false, but that it described one case rather than the general one. The model assumed users who could not talk, could not watch, and could not bind themselves, and for such users the grim conclusion does follow. Real communities are frequently able to do all three, and the capacity to make and enforce local agreements is itself a resource, one that erodes if outsiders, however well intentioned, override it. A government that replaces a working local arrangement with a uniform national rule may extinguish the very monitoring and trust on which restraint depended, and so produce the tragedy it meant to prevent.

Q30. [MBA2-VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the central argument of the passage?

- A) Communities can sometimes govern shared resources by self-made rules, so the pessimistic model fits a special case ✓
- B) State regulation from above is shown to be the single most dependable remedy for the overuse of any common resource
- C) Traditional rural communities, unlike modern states, essentially never deplete the natural resources placed in their care
- D) Shared resources are inevitably destroyed unless they are divided up into privately owned parcels by their own users

Answer: A) Communities can sometimes govern shared resources by self-made rules, so the pessimistic model fits a special case

Solution: The passage argues the grim model holds only for users who cannot monitor, communicate, or bind themselves, while many real communities can and so govern their commons without nationalising or privatising them. The privatisation-inevitable and regulation-most-dependable options each pick one of the two paths the passage relativises. The 'never deplete' option overstates: the text says the prediction 'often held' where the enabling conditions were absent.

Q31. [MBA2-VARC-RC2 · mcsc · diff 0.5]

The passage credits the durable commons arrangements with all of the following features EXCEPT:

- A) provisions allowing the governing rules to be revised as surrounding conditions change
- B) penalties that fall lightly on a first offence and grow heavier when the offence is repeated
- C) the ability of users to keep watch on one another and to communicate about the resource
- D) the assignment of a separately owned private parcel of the resource to each individual user ✓

Answer: D) the assignment of a separately owned private parcel of the resource to each individual user

Solution: The enduring commons were explicitly neither privatised nor nationalised, so assigning each user a private parcel is the feature the passage denies and is the EXCEPT. Graduated penalties, mutual monitoring and communication, and revisable rules are each named directly as pillars of the successful arrangements.

Q32. [MBA2-VARC-RC2 · mcsc · diff 1]

The author observes that a well-meaning government can 'produce the tragedy it meant to prevent' mainly in order to:

- A) warn that overriding a working local arrangement can destroy the conditions that sustained restraint ✓
- B) argue that governments are, almost as a rule, indifferent to whether shared resources survive or collapse
- C) claim that locally made rules are plainly superior to state rules in virtually every respect and setting
- D) show that dividing a resource into private holdings will reliably outperform any centrally imposed public rule

Answer: A) warn that overriding a working local arrangement can destroy the conditions that sustained restraint

Solution: The point is that a uniform national rule can extinguish the monitoring and trust on which local restraint depended, so intervention can backfire. It does not impute indifference to governments, and it does not claim local rules win everywhere: the text stresses local success is conditional on monitoring and communication. Privatisation is not endorsed here at all.

Q33. [MBA2-VARC-RC2 · mcsc · diff 1]

The author's stance toward the original pessimistic model is best described as:

- A) full endorsement of it as a fixed law that governs shared resources in every setting
- B) studied indifference, on the ground that the model's assumptions are said to be untestable
- C) outright rejection of it as an account that field evidence has now proved plainly false
- D) qualified acceptance, treating it as accurate for a restricted range of conditions ✓

Answer: D) qualified acceptance, treating it as accurate for a restricted range of conditions

Solution: The author says explicitly the lesson is 'not that the original model was false' but that it fits one case, and that for users who cannot talk, watch, or bind themselves 'the grim conclusion does follow'. That is qualified acceptance. Outright rejection contradicts the disclaimer; full endorsement ignores the restriction to a special case; indifference is wrong because the model is engaged with closely and its assumptions are tested against field cases.

Q34. [MBA2-VARC-ODD-1 · mcsc · diff 0.5]

Five sentences are given below. Four of them combine to form a single coherent paragraph; one does not belong. Read the sentences and identify the misfit.

- (1) A tidal wetland acts as a natural buffer, absorbing the energy of incoming storm surges before they reach inland settlements.
- (2) Its dense mat of roots binds loose sediment, so the shoreline erodes far more slowly than an unvegetated coast would.
- (3) The same roots filter runoff, trapping pollutants that would otherwise flow straight into open water.
- (4) Because of these functions, planners increasingly treat wetlands as infrastructure rather than as idle land.
- (5) Modern desalination plants can convert seawater into drinking water at a steadily falling cost per litre.

Which sentence is the odd one out?

- A) (3)
- B) (4)
- C) (5) ✓
- D) (2)
- E) (1)

Answer: C) (5)

Solution: Sentences 1, 2, 3 and 4 build one argument: wetlands buffer surges, hold sediment, filter runoff, and so count as infrastructure. The desalination-plant sentence introduces an unrelated technology for producing drinking water and shares none of that thread, so it is the intruder. Sentence 4 begins with 'these functions', which points back to the wetland benefits, confirming the other four cohere.

Q35. [MBA2-VARC-ODD-2 · mcsc · diff 0.5]

Five sentences are given below. Four of them combine to form a single coherent paragraph; one does not belong. Read the sentences and identify the misfit.

- (1) When a folk tale passes from one teller to the next, small changes creep in with every retelling.
- (2) A teller may drop a name, sharpen a joke, or shift the setting to a place the listeners know.
- (3) Over generations these tiny edits accumulate into strikingly different versions of one story.
- (4) Scholars can sometimes trace those versions back to a shared ancestor by comparing their fixed motifs.
- (5) Printing presses lowered the price of books so sharply that private libraries became common.

Which sentence is the odd one out?

- A) (2)
- B) (1)
- C) (3)
- D) (5) ✓
- E) (4)

Answer: D) (5)

Solution: The four connected sentences describe how oral folk tales mutate as they are retold and how scholars reconstruct their common origin. The sentence about printing presses and the price of books concerns written distribution, a different subject that breaks the chain about oral transmission, making it the misfit. The word 'those versions' in sentence 4 links back to the accumulated edits, tying the other four together.

Q36. [MBA2-VARC-ODD-3 · mcsc · diff 1]

Five sentences are given below. Four of them combine to form a single coherent paragraph; one does not belong. Read the sentences and identify the misfit.

- (1) A scientific model is deliberately simpler than the reality it represents, keeping only the features that matter for a given question.
- (2) By stripping away detail, it lets researchers reason about a system that would otherwise be too tangled to follow.
- (3) The value of a model therefore lies not in its completeness but in the useful predictions it yields.
- (4) A model that reproduced every detail of the world would be as hard to understand as the world itself.
- (5) A well-designed model can be tested against fresh observations and discarded the moment it fails to predict them.

Which sentence is the odd one out?

- A) (5) ✓
- B) (2)
- C) (1)
- D) (3)
- E) (4)

Answer: A) (5)

Solution: Sentences 1, 2, 3 and 4 develop one idea, that a model earns its worth through simplification rather than through completeness. Sentence 5 is subtly off: it stays on the topic of models but shifts to a different claim, about falsification and testing against new data, which the other four never raise. Because 5 introduces a separate virtue instead of extending the completeness-versus-usefulness argument, it is the odd one out.

Q37. [MBA2-VARC-ODD-4 · mcsc · diff 1]

Five sentences are given below. Four of them combine to form a single coherent paragraph; one does not belong. Read the sentences and identify the misfit.

- (1) A currency holds its value only because the people using it expect others to accept it tomorrow.
- (2) That expectation is a shared belief, sustained by habit and by trust in the issuing authority.
- (3) When the belief weakens, holders rush to spend, and prices climb faster than the notes can be printed.
- (4) A sudden loss of confidence can thus unravel a currency long before any change in the goods it can buy.
- (5) Central banks set interest rates to steer how much borrowing and lending take place across the economy.

Which sentence is the odd one out?

- A) (4)
- B) (5) ✓
- C) (2)
- D) (1)
- E) (3)

Answer: B) (5)

Solution: The cohering four trace one causal story: a currency's value rests on shared expectation, that expectation is belief, weakened belief triggers a spending rush, and confidence can collapse before real goods change. Sentence 5 also concerns central banking but makes a different claim, about interest-rate policy and credit, rather than the psychology of confidence. It sits in the same field yet breaks the single argument, so it is the intruder.

Q38. [MBA2-VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. Every autumn, millions of songbirds leave the cooling north for warmer southern feeding grounds.
2. To survive the journey, each bird nearly doubles its body weight by gorging on insects and berries beforehand.
3. This stored fat becomes the fuel that powers days of almost continuous flight.
4. Without it, a small migrant could not cross the wide deserts and seas that lie along the route.

Select the correct order.

- A) 2134
- B) 1324
- C) 1243
- D) 1234 ✓

Answer: D) 1234

Solution: Sentence 1 sets the scene with the autumn departure and is the natural opener. Sentence 2 explains the preparation, doubling body weight, which sentence 3 then labels 'this stored fat' and calls the fuel. Sentence 4 closes with 'Without it', referring to that fat, and names the obstacles. The chain 1-2-3-4 keeps each pronoun next to its referent, so the order is 1234.

Q39. [MBA2-VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. A public library began, in most towns, as a single donated room lined with second-hand books.
2. As demand grew, the room acquired a paid keeper who lent volumes and tracked their return.
3. That keeper slowly became a trained profession, with rules for cataloguing and preserving the collection.
4. Today those same rules underpin vast digital archives that no single room could ever hold.

Select the correct order.

- A) 1324
- B) 1243
- C) 2134
- D) 1234 ✓

Answer: D) 1234

Solution: The paragraph moves through time. Sentence 1 gives the humble origin, a donated room. Sentence 2 adds the paid keeper as demand grows, and sentence 3 turns 'that keeper' into a profession with rules. Sentence 4 begins 'Today those same rules', extending the cataloguing rules to digital archives. Each step depends on the one before, so the sequence is 1234.

Q40. [MBA2-VARC-JUMBLE-3 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. Critics once dismissed the invention as a toy with no serious use.
2. Its defenders replied that the same had been said of the telephone in its first decade.
3. They pointed out that such judgements track familiarity, not the real limits of a device.
4. History, they argued, rarely rewards those who mistake a rough prototype for a finished tool.

Select the correct order.

- A) 1324
- B) 1234 ✓
- C) 2134
- D) 1243

Answer: B) 1234

Solution: Sentence 1 states the dismissal and must open. Sentence 2 begins 'Its defenders replied', a direct rebuttal that has to follow the criticism. Sentence 3 opens with 'They', the defenders, generalising their reply, and sentence 4 continues 'they argued' with the closing lesson. Reordering breaks the pronoun 'They', which needs 'defenders' named first, so only 1234 keeps the references intact.

Q41. [MBA2-VARC-JUMBLE-4 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. A map must leave things out, for a chart that showed every detail would be useless.
2. Yet what it leaves out is never neutral, since each omission quietly reflects the mapmaker's priorities.
3. Such choices, repeated across thousands of maps, shape how a whole society pictures its own territory.
4. For this reason a map is best read not only as a guide but as an argument about what deserves notice.

Select the correct order.

- A) 2134
- B) 1324
- C) 1234 ✓
- D) 1243

Answer: C) 1234

Solution: Sentence 1 introduces the necessary omission and opens the paragraph. Sentence 2 begins 'Yet what it leaves out', a contrast that must follow the claim about leaving things out. Sentence 3 refers to 'Such choices', the priority-driven omissions of sentence 2, and sentence 4 draws the conclusion with 'For this reason'. The connectives Yet, Such choices, and For this reason force the order 1234.

Q42. [MBA2-VARC-JUMBLE-5 · mcsc · diff 1]

The four sentences below, labelled 1 to 4, have been jumbled. Arrange them to form a coherent paragraph.

1. Silence in a conversation is often read as a gap, an awkward failure to keep talking.
2. In many cultures, however, it carries meaning as rich as any spoken reply.
3. There a considered pause signals respect, telling the other that the question deserved real thought.
4. To fill that pause too quickly can therefore seem not friendly but careless.

Select the correct order.

- A) 1243
- B) 1234 ✓
- C) 2134
- D) 1324

Answer: B) 1234

Solution: Sentence 1 sets up the common view of silence as an awkward gap. Sentence 2 answers with 'however', reframing silence as meaningful, and sentence 3 begins 'There', pointing to those cultures and explaining what the pause signals. Sentence 4 concludes with 'that pause' and 'therefore', drawing the consequence. The linking words however, There, and therefore fix the reading order as 1234.

Q43. [MBA2-VARC-SUMMARY-1 · mcsc · diff 0.5]

Many species that appear wasteful are in fact insuring themselves against bad years. An oak may drop thousands of acorns although only a handful can ever grow into trees, because the surplus swamps the animals that eat them and guarantees that a few seeds survive in almost any season. What looks like extravagance is a hedge: by producing far more than the average conditions require, the tree buys protection against the years when conditions are worst.

Which option best summarises the paragraph?

A) Apparent overproduction in nature is often a safeguard, letting a species survive poor years by making far more offspring than good conditions alone would need. ✓

B) Animals that feed on seeds are usually overwhelmed by the sheer number a plant produces and so fail to eat the entire crop.

C) Oak trees drop many thousands of acorns every single year even though almost none of those scattered acorns will ever manage to grow into a mature standing tree.

D) Nature tends to be wasteful, and much of what living things produce serves no real purpose beyond simple excess.

Answer: A) Apparent overproduction in nature is often a safeguard, letting a species survive poor years by making far more offspring than good conditions alone would need.

Solution: The passage argues that seeming waste is really insurance against bad years, using the oak only as an example. The correct summary captures that general point about overproduction as a safeguard. The acorn statement and the seed-eater statement are single supporting details, too narrow to be the summary, and the 'nature is wasteful' option flatly contradicts the passage, which says the surplus has a clear purpose.

Q44. [MBA2-VARC-SUMMARY-2 · mcsc · diff 0.5]

A new road built to relieve congestion often ends up just as crowded as before. The extra capacity lowers the effective cost of driving, so people who once avoided the trip, took the train, or travelled at odd hours now join the traffic. Within a few years the freed-up space fills again, and the promised relief quietly disappears even though the road itself was built exactly as planned.

Which option best summarises the paragraph?

A) Roads are frequently built exactly to their original plans, yet the traffic engineers still misjudge how drivers will respond.

B) Adding road capacity tends to attract new trips that fill the space, so the expected easing of congestion often fails to last. ✓

C) Traffic congestion is one of the most stubborn and expensive problems that fast-growing modern cities are forced to confront.

D) People who used to take the train will switch to driving as soon as a wider and more convenient road becomes available to them.

Answer: B) Adding road capacity tends to attract new trips that fill the space, so the expected easing of congestion often fails to last.

Solution: The paragraph's core claim is that new capacity induces extra demand that refills the road, defeating the intended relief. The correct option states exactly that cause and effect. The train-switching option is only one instance of the induced demand, the 'built to plan' option seizes a side remark, and the 'congestion is stubborn' option is a general theme the passage never actually argues.

Q45. [MBA2-VARC-SUMMARY-3 · mcsc · diff 0.5]

Learning a second language as an adult rarely produces a flawless accent, yet the effort is far from wasted. Even partial fluency reshapes how the learner hears speech, sharpens attention to grammar in the first language, and opens doors to texts and people that were closed before. The goal need not be to pass as a native; the practical and mental gains arrive long before that distant point.

Which option best summarises the paragraph?

- A) Studying grammar in a new language happens to sharpen a learner's awareness of the grammar of their own first language.
- B) Reaching the fluent level of a genuine native speaker is really the one true and proper measure of success for anyone setting out to learn a new foreign language.
- C) Adult language learning brings real practical and mental benefits well before fluency, so a perfect accent need not be the aim. ✓
- D) Adults who study a second language almost never manage to acquire an accent that a native speaker would accept as flawless.

Answer: C) Adult language learning brings real practical and mental benefits well before fluency, so a perfect accent need not be the aim.

Solution: The passage says adult learners seldom sound native but gain much sooner, so a perfect accent is not the point. The correct summary states both halves, benefits arrive early and native-level need not be the goal. The accent option and the grammar option are isolated details, and the 'native speaker is the true measure' option reverses the passage's actual conclusion.

Q46. [MBA2-VARC-SUMMARY-4 · mcsc · diff 1]

Rating systems promise to guide us toward quality, but a high average can mislead as easily as it informs. A product reviewed by only a dozen buyers may score higher than a rival reviewed by thousands, not because it is better but because a small sample swings more wildly. Worse, the people who bother to rate are rarely typical: the delighted and the furious speak up while the merely satisfied stay silent. A score, then, summarises not the product alone but the peculiar crowd that chose to judge it, and reading it wisely means asking who did the rating before trusting what they said.

Which option best summarises the paragraph?

- A) A product judged by only a handful of reviewers can post a higher average than a rival that thousands of buyers have carefully rated.
- B) Rating systems were originally designed to steer ordinary shoppers toward higher quality goods, and for the most part they seem to carry out that useful task quite reliably.
- C) People who feel delighted or furious about a purchase are much more likely to leave a review than those who feel merely satisfied.
- D) Average ratings can mislead because small samples fluctuate and reviewers are unrepresentative, so a score reflects who rated as much as the product. ✓

Answer: D) Average ratings can mislead because small samples fluctuate and reviewers are unrepresentative, so a score reflects who rated as much as the product.

Solution: The passage makes two linked points, that small samples swing and that reviewers are a skewed crowd, concluding a score reflects its raters. Only the correct option carries that full scope. The small-sample option and the delighted-or-furious option are each one of the two reasons, true but partial, and the reliability option contradicts the passage by calling ratings reliable, which is the very claim being questioned.

Q47. [MBA2-VARC-SUMMARY-5 · mcsc · diff 1]

We tend to praise the lone inventor, but most breakthroughs are less a single leap than a last step. By the time a discovery is announced, dozens of others have usually reached the same threshold, held back only by a missing tool or an unasked question. The celebrated name is often simply the one who arrived days ahead, or who framed the result in words the age was ready to hear. To credit a single mind, then, is to mistake the visible tip of a long collective effort for the whole of it.

Which option best summarises the paragraph?

A) The lone individual inventor still fully deserves the praise that we give, because framing a raw result in exactly the right words is itself the single hardest part of any discovery.

B) Great discoveries are usually the final step of a wide shared effort, so crediting one inventor mistakes a visible moment for the whole. ✓

C) A breakthrough is frequently delayed by nothing more than a missing instrument or a question that no one has yet thought to ask.

D) Whoever announces a discovery first has often arrived only a few days ahead of several rivals working on the very same problem.

Answer: B) Great discoveries are usually the final step of a wide shared effort, so crediting one inventor mistakes a visible moment for the whole.

Solution: The passage argues that breakthroughs cap a broad collective effort and that honouring one mind mistakes the tip for the whole. The correct option states both the shared-effort claim and its consequence. The 'arrived days ahead' and 'missing instrument' options are supporting details, and the final option endorses the lone-inventor view the passage sets out to overturn.

Q48. [MBA2-VARC-SUMMARY-6 · mcsc · diff 1]

A law that everyone quietly ignores can be more corrosive than no law at all. When a rule is kept on the books but never enforced, it does not simply lie dormant; it teaches citizens that the written word and the real world need not match. Each unpunished breach makes the next one easier to excuse, and over time the habit of selective obedience spreads to rules that were meant to bind. The damage lies less in the single ignored statute than in the lesson it spreads about how seriously any law should be taken.

Which option best summarises the paragraph?

A) Once one violation of a statute goes unpunished, excusing the next similar violation becomes noticeably easier to justify.

B) An unenforced law erodes respect for law in general, because tolerating its breach teaches that rules need not be obeyed. ✓

C) Lawmakers should take care to repeal statutes they no longer intend to enforce rather than leaving them on the books.

D) A rule that stays on the books without ever being enforced does not vanish but quietly remains available for later use.

Answer: B) An unenforced law erodes respect for law in general, because tolerating its breach teaches that rules need not be obeyed.

Solution: The passage's thesis is that an ignored law spreads a corrosive lesson, weakening obedience to law as a whole. The correct option captures that broad erosion and its cause. The 'remains available' option distorts a minor phrase, the 'next violation easier' option is one step in the argument rather than the whole, and the repeal recommendation is advice the passage never offers.

Section: DILR (40 min)

Instructions (MBA1-DILR-1): study the shared data below and answer its 4 question(s).

Five inspectors (Lena, Marco, Nadia, Omar and Priya) each audited exactly one of five sites (Alpha, Birch, Cedar, Dune and Elm) during a single working week. Every inspector worked on a different weekday from Monday to Friday, and no site was visited twice. Each audit received a distinct compliance score from the list 12, 15, 18, 21 and 24.

- Lena's audit took place before Marco's, and Marco's took place before Omar's. - Cedar was audited on Wednesday. - Nadia audited Alpha, on the day immediately after Priya's audit. - Elm was audited earlier in the week than Birch, and Dune was audited on Friday. - Lena did not audit Cedar. - The Elm audit scored 18, and Marco's score was exactly 9 more than Nadia's. - Priya's score was lower than both Nadia's score and Omar's score.

Q49. [MBA1-DILR-1 · mcsc · diff 1]

Which inspector audited Birch?

- A) Priya
- B) Nadia
- C) Lena
- D) Marco ✓

Answer: D) Marco

Solution: Dune is Friday and Cedar is Wednesday, so the Nadia-immediately-after-Priya pair must sit inside Monday to Thursday. With Lena before Marco before Omar, Omar is pushed to Friday (Dune) and the only consistent fit is Lena Monday, Marco Tuesday, Priya Wednesday, Nadia Thursday. Since Nadia has Alpha, Priya has Cedar and Elm must precede Birch, Lena takes Elm on Monday and Marco takes Birch on Tuesday. Naming Priya or Nadia ignores that their sites are already fixed as Cedar and Alpha, and naming Lena ignores the Elm-before-Birch order.

Q50. [MBA1-DILR-1 · mcsc · diff 1]

What is the combined compliance score of the Monday audit and the Friday audit?

- A) 42
- B) 33
- C) 30
- D) 39 ✓

Answer: D) 39

Solution: The schedule is Lena Monday Elm, Marco Tuesday Birch, Priya Wednesday Cedar, Nadia Thursday Alpha, Omar Friday Dune. Elm scored 18, so Lena has 18. Marco being 9 above Nadia leaves the pairs 24 with 15 or 21 with 12; the second pair forces Nadia to 12, which contradicts Priya being lower than Nadia, so Marco has 24 and Nadia 15. Priya then takes 12 and Omar 21, giving 18 plus 21 equals 39. The value 42 comes from pairing 18 with 24, 33 from using Nadia's 15 and 30 from using Priya's 12.

Q51. [MBA1-DILR-1 · mcsc · diff 1]

Which of the following statements is true?

- A) Priya recorded the lowest of the five scores and worked on Wednesday. ✓
- B) Lena audited Elm and scored higher than Marco did on the same week.
- C) Omar recorded the highest of the five scores, and he worked on Friday.
- D) Nadia audited Alpha on the day immediately before Cedar was audited.

Answer: A) Priya recorded the lowest of the five scores and worked on Wednesday.

Solution: The unique solution is Lena Monday Elm 18, Marco Tuesday Birch 24, Priya Wednesday Cedar 12, Nadia Thursday Alpha 15, Omar Friday Dune 21. Priya's 12 is indeed the lowest score and her day is Wednesday. Omar does work on Friday but 21 is not the highest score, since Marco has 24. Nadia's Thursday comes after, not before, the Wednesday Cedar audit. Lena did audit Elm but her 18 is below Marco's 24.

Q52. [MBA1-DILR-1 · mcsc · diff 1]

Suppose the information that Dune was audited on Friday is replaced by the information that Dune was audited on Thursday, with every other statement unchanged. Which site is then audited on Friday?

- A) Cedar
- B) Birch
- C) Dune
- D) Alpha ✓

Answer: D) Alpha

Solution: With Dune on Thursday and Cedar on Wednesday, the consecutive Priya then Nadia pair can only be Thursday then Friday, so Nadia takes Friday with Alpha. Lena before Marco before Omar then fills Monday, Tuesday and Wednesday in that order, and Elm before Birch forces Lena on Elm and Marco on Birch, leaving Omar with Cedar on Wednesday and Priya with Dune on Thursday. Cedar and Dune are ruled out by their stated days, and Birch is locked to Tuesday by the ordering.

Instructions (MBA1-DILR-2): study the shared data below and answer its 3 question(s).

Six crates labelled K, L, M, N, O and R are stacked one above another in a single tower. Position 1 is the bottom of the tower and position 6 is the top, and each position holds exactly one crate.

- Crate M rests immediately above crate K. - Exactly two crates lie between L and M. - R is neither at the bottom nor at the top of the tower. - O is somewhere above L. - Exactly one crate lies between N and R.

Q53. [MBA1-DILR-2 · mcsc · diff 0.5]

Which crate is at the bottom of the tower?

- A) K
- B) L
- C) N ✓
- D) R

Answer: C) N

Solution: The K then M block plus two crates between L and M leaves only a few layouts, and requiring exactly one crate between N and R while R avoids both ends fixes the tower as N, L, R, K, M, O from bottom to top. So N is at the bottom. L sits second, K fourth and R is barred from position 1 outright by the end restriction.

Q54. [MBA1-DILR-2 · mcsc · diff 0.5]

How many crates rest above R?

- A) 3 ✓
- B) 4
- C) 2
- D) 1

Answer: A) 3

Solution: The tower is N, L, R, K, M, O from bottom to top, so R occupies position 3 and the crates K, M and O lie above it, giving 3. Answering 2 counts only the K and M block, 4 comes from placing R at position 2 and 1 comes from reading the stack top down instead of bottom up.

Q55. [MBA1-DILR-2 · mcsc · diff 1]

If the condition that O is somewhere above L is dropped and all the other conditions still hold, how many different stacking orders are possible?

- A) 3
- B) 4 ✓
- C) 2
- D) 5

Answer: B) 4

Solution: Without the O above L condition, the K then M block can sit at positions 1 and 2, 4 and 5 or 5 and 6 while keeping two crates between L and M, R off both ends and exactly one crate between N and R. Checking each case gives K M O R L N, N L R K M O, O N L R K M and O R L N K M, so 4 orders survive. Answering 3 misses the case with the block at the very bottom, while 5 and 2 come from mishandling the N and R gap.

Instructions (MBA1-DILR-3): study the shared data below and answer its 4 question(s).

An audit of four solar module assemblers covers three panel grades, G1, G2 and G3. Table 1 gives the number of modules installed and Table 2 gives the share of those modules found defective in the audit.

Table 1: Modules installed (in hundreds)

Firm	G1	G2	G3
Arkin	24	36	20
Bhima	30	18	42
Cygna	45	25	15
Dhruv	16	40	34

Table 2: Defective modules as a percentage of modules installed

Firm	G1	G2	G3
Arkin	2	5	4
Bhima	6	3	2
Cygna	4	2	8
Dhruv	5	4	3

Q56. [MBA1-DILR-3 · mcsc · diff 1]

In grade G3, how many more defective modules did Cygna and Dhruv together have than Arkin and Bhima together?

- A) 46
- B) 64
- C) 58 ✓
- D) 38

Answer: C) 58

Solution: Defective modules equal installed hundreds times the rate, so each product is (hundreds x rate). G3 gives Arkin $20 \times 4 = 80$, Bhima $42 \times 2 = 84$, Cygna $15 \times 8 = 120$, Dhruv $34 \times 3 = 102$. So 222 against 164, a gap of 58. The 46 trap swaps two rate entries, 64 comes from reading G2 rates, and 38 subtracts firm totals rather than G3 alone.

Q57. [MBA1-DILR-3 · mcsc · diff 1]

Taking all four firms and all three grades together, defective modules form approximately what percentage of all modules installed?

- A) 3.82% ✓
- B) 3.90%
- C) 4.05%
- D) 3.68%

Answer: A) 3.82%

Solution: Total installed is $(80+90+85+90)$ hundreds = 34500 modules. Total defective is $308+318+350+342 = 1318$, so the overall rate is $1318/34500 = 3.82$ percent. The 3.90 and 4.05 values come from averaging the twelve percentages instead of weighting them by volume, and 3.68 drops one grade column from the defective total.

Q58. [MBA1-DILR-3 · mcsc · diff 1]

Which firm has the lowest defect rate across its entire installed base?

- A) Dhruv
- B) Bhima ✓
- C) Cygna
- D) Arkin

Answer: B) Bhima

Solution: Overall rates must be volume weighted: Arkin $308/8000 = 3.85$ percent, Bhima $318/9000 = 3.53$ percent, Cygna $350/8500 = 4.12$ percent, Dhruv $342/9000 = 3.80$ percent. Bhima is lowest despite carrying the single worst cell rate of 6 percent, because that rate sits on its smallest grade mix. Dhruv and Arkin look lower if the three grade rates are simply averaged.

Q59. [MBA1-DILR-3 · mcsc · diff 1]

By how many modules does the firm with the largest number of defective modules exceed the firm with the smallest?

- A) 24
- B) 42 ✓
- C) 34
- D) 56

Answer: B) 42

Solution: Firm defective totals are Arkin 308, Bhima 318, Cygna 350 and Dhruv 342. The largest is Cygna at 350 and the smallest is Arkin at 308, a difference of 42. The 34 value compares Bhima with Arkin, 24 compares Dhruv with Bhima, and 56 comes from using installed volumes rather than defective counts.

Instructions (MBA1-DILR-4): study the shared data below and answer its 4 question(s).

A depot dispatched exactly 48 crates of packaged tea to four towns, Alwar, Betul, Chirala and Dhanbad, on a single morning. The crates for each town went on that town's own truck, and the four trucks left the depot one after another, first to fourth.

- Each town received a distinct number of crates, and every one of those numbers was a multiple of 4 lying between 4 and 24, both inclusive.
- Chirala received more crates than Alwar.
- The town whose truck left first received exactly twice as many crates as the town whose truck left fourth.
- Dhanbad's truck left immediately after Betul's truck.
- Alwar's truck did not leave first, and Alwar received fewer crates than Betul.
- The town whose truck left third received exactly 12 crates.
- Chirala's truck left before Betul's truck.

Q60. [MBA1-DILR-4 · mcsc · diff 0.5]

How many crates did Chirala receive?

- A) 12 crates
- B) 24 crates
- C) 4 crates
- D) 8 crates ✓

Answer: D) 8 crates

Solution: Chirala leaves before Betul and Alwar is not first, while Dhanbad follows Betul immediately, so the order can only be Chirala, Betul, Dhanbad, Alwar or Chirala, Alwar, Betul, Dhanbad. The third truck carries 12 crates and the first carries twice the fourth. Testing distinct multiples of 4 from 4 to 24 that total 48 leaves only Chirala 8, Betul 24, Dhanbad 12, Alwar 4 in the order Chirala, Betul, Dhanbad, Alwar. Hence Chirala received 8 crates. The 4 figure is Alwar's share, 12 is Dhanbad's (the third truck), and 24 is Betul's, the largest allocation.

Q61. [MBA1-DILR-4 · mcsc · diff 1]

Which town's truck left second, and how many crates did that town receive?

- A) Chirala, with 8 crates
- B) Betul, with 24 crates ✓
- C) Alwar, with 20 crates
- D) Dhanbad, with 12 crates

Answer: B) Betul, with 24 crates

Solution: The only allocation consistent with every clue is Chirala 8, Betul 24, Dhanbad 12, Alwar 4, dispatched in the order Chirala, Betul, Dhanbad, Alwar. So the second truck was Betul's, carrying 24 crates. Dhanbad with 12 crates describes the third truck, Chirala with 8 the first, and Alwar never received 20 crates in any consistent arrangement since 24 is fixed for Betul.

Q62. [MBA1-DILR-4 · mcsc · diff 1]

If the clue that the third truck's town received exactly 12 crates is withdrawn and all the other clues still hold, how many complete arrangements of dispatch order and crate counts are possible?

- A) 7 arrangements ✓
- B) 6 arrangements
- C) 9 arrangements
- D) 5 arrangements

Answer: A) 7 arrangements

Solution: Without the 12-crate clue the order is still either Chirala, Betul, Dhanbad, Alwar or Chirala, Alwar, Betul, Dhanbad. For the first order the possibilities are Alwar 4 with Betul and Dhanbad taking 12 and 24, 16 and 20, 20 and 16 or 24 and 12 alongside Chirala 8, plus Alwar 8 with Chirala 16, Betul 20, Dhanbad 4, which is five cases; the second order adds two more, giving 7 in all. The 5 count stops at the first order only, 6 misses one of the extra cases, and 9 double counts allocations that repeat a crate figure.

Q63. [MBA1-DILR-4 · mcsc · diff 1]

Suppose the first truck's town received three times, rather than twice, as many crates as the fourth truck's town, every other clue remaining as stated. How many crates would Dhanbad then receive?

- A) 8 crates ✓
- B) 24 crates
- C) 4 crates
- D) 12 crates

Answer: A) 8 crates

Solution: Under the revised ratio the arrangement becomes Chirala 24, Alwar 4, Betul 12, Dhanbad 8, dispatched in that order: the third truck is Betul's with 12 crates, the first is Chirala's with 24, which is three times Alwar's 4 on the fourth truck, and the four counts still total 48. So Dhanbad receives 8 crates. The 4 belongs to Alwar, 12 to the third truck Betul, and 24 to Chirala, whose figure is often read off as Dhanbad's by mistake.

Instructions (MBA1-DILR-5): study the shared data below and answer its 3 question(s).

A training coordinator must schedule exactly four of seven half-day workshops, namely Auditing, Branding, Coding, Design, Ethics, Finance and Grammar, for the coming quarter.

- Coding is scheduled only if Design is also scheduled.
- Branding and Ethics are run by the same trainer, so they cannot both be scheduled.
- Auditing and Finance need the same laboratory, so they cannot both be scheduled.
- Grammar is scheduled if and only if Auditing is scheduled.
- If Design is scheduled, then Branding is not scheduled.

Q64. [MBA1-DILR-5 · mcsc · diff 0.5]

How many different groups of four workshops can the coordinator schedule?

- A) 3 groups ✓
- B) 4 groups
- C) 5 groups
- D) 2 groups

Answer: A) 3 groups

Solution: Checking all 35 groups of four against the conditions leaves only Auditing, Coding, Design and Grammar; Auditing, Design, Ethics and Grammar; and Coding, Design, Ethics and Finance. That is 3 groups. The 2 count comes from overlooking the group that omits Ethics, while 4 and 5 come from admitting groups that break either the Grammar and Auditing pairing or the ban on Design with Branding.

Q65. [MBA1-DILR-5 · mcsc · diff 0.5]

Which workshop cannot appear in any acceptable group of four?

- A) Grammar
- B) Branding ✓
- C) Auditing
- D) Coding

Answer: B) Branding

Solution: If Branding is scheduled then Ethics and Design are both out, so the four must be drawn from Auditing, Coding, Finance and Grammar; but Coding needs Design, Auditing and Finance clash, and Grammar needs Auditing, so no such group exists. Branding therefore never appears. Auditing and Grammar appear together in two acceptable groups, and Coding appears in two acceptable groups as well.

Q66. [MBA1-DILR-5 · mcsc · diff 1]

If Finance is scheduled, which workshops make up the rest of the group?

- A) Coding, Design and Grammar
- B) Auditing, Design and Grammar
- C) Design, Ethics and Grammar
- D) Coding, Design and Ethics ✓

Answer: D) Coding, Design and Ethics

Solution: Finance rules out Auditing, and with Auditing out Grammar is out too, so the remaining three come from Branding, Coding, Design and Ethics. Branding cannot join Design, and dropping Design would force Coding out as well, leaving too few workshops, so Design stays, Branding goes, and Coding and Ethics complete the group: Coding, Design and Ethics. Any option containing Grammar fails because Grammar travels only with Auditing, which Finance excludes.

Instructions (MBA1-DILR-6): study the shared data below and answer its 4 question(s).

An industrial electronics maker reports units shipped and the average price realised per unit for its three product lines over three years. Revenue for a line equals units shipped multiplied by average price.

Q67. [MBA1-DILR-6 · mcsc · diff 1]

What was the total revenue of the three lines together in 2024, in Rs lakh?

- A) 2612
- B) 2537 ✓
- C) 2350
- D) 2465

Answer: B) 2537

Solution: Revenue in Rs lakh is units in thousands times price divided by 100. For 2024 that gives Sensors $55 \times 1400 / 100 = 770$, Controllers $27 \times 2500 / 100 = 675$ and Drives $21 \times 5200 / 100 = 1092$, summing to 2537. The 2465 and 2350 values carry over a 2023 price or volume, and 2612 uses the 2023 Drives price of 4800 against the wrong volume.

Q68. [MBA1-DILR-6 · mcsc · diff 1]

The line with the highest percentage growth in revenue from 2022 to 2024 grew by approximately what percentage?

- A) 92%
- B) 60%
- C) 82% ✓
- D) 71%

Answer: C) 82%

Solution: Revenues in Rs lakh move from Sensors 480 to 770, Controllers 600 to 675 and Drives 600 to 1092. The growth rates are about 60 percent, 12.5 percent and 82 percent, so Drives leads at 82 percent. The 60 percent figure belongs to Sensors, 71 percent comes from growing only the volume of Drives from 12 to 21 read against price, and 92 percent overstates the base as 570.

Q69. [MBA1-DILR-6 · mcsc · diff 1]

In 2023, what was the average price per unit realised across all units shipped, in Rs?

- A) 2280 ✓
- B) 2124
- C) 2217
- D) 2350

Answer: A) 2280

Solution: Total 2023 revenue is $575+780+720 = 2075$ Rs lakh on $46+30+15 = 91$ thousand units, so the blended price is $207500000/91000$, about Rs 2280. The 2217 value is the plain average of 1250, 2600 and 4800 scaled wrongly, 2350 mixes in 2024 prices, and 2124 uses 2022 volumes as weights instead of 2023 volumes.

Q70. [MBA1-DILR-6 · mcsc · diff 1]

By approximately what percentage did the combined revenue of the three lines change from 2023 to 2024?

- A) 18.2%
- B) 22.3% ✓
- C) 24.9%
- D) 20.5%

Answer: B) 22.3%

Solution: Combined revenue is $575+780+720 = 2075$ Rs lakh in 2023 and $770+675+1092 = 2537$ Rs lakh in 2024, an increase of 462 on 2075, that is 22.3 percent. The 18.2 percent value divides the rise by the 2024 total, 24.9 percent averages the three line growth rates, and 20.5 percent tracks units alone while ignoring the price changes.

Instructions (MBA2-DILR-1): study the shared data below and answer its 4 question(s).

Five employees, Mira, Omar, Priya, Rehan and Sana, sit at a row of five desks numbered 1 to 5 from left to right. Each works in exactly one of five departments (Finance, Legal, Sales, Tech, HR), and each has a distinct number of years of experience drawn from 2, 4, 6, 8 and 10.

- The Finance employee sits immediately to the left of the Legal employee. - Priya has more experience than exactly three of her colleagues. - The Tech employee has the least experience. - Omar sits at desk 3 and works in Finance. - Sana has exactly 4 years more experience than the Sales employee, and Sana does not work in Sales. - The employee at desk 1 has 6 years of experience. - Mira sits somewhere to the right of Rehan. - The HR employee sits at desk 5. - The Legal employee has 8 years of experience.

Q71. [MBA2-DILR-1 · mcsc · diff 1]

Which employee sits at desk 4?

- A) Sana
- B) Mira
- C) Priya ✓
- D) Omar

Answer: C) Priya

Solution: Priya has more experience than exactly three others, so she holds 8 years, and the Legal person also holds 8, making Priya the Legal employee. Omar is Finance at desk 3, so Legal sits immediately to his right at desk 4. Hence Priya is at desk 4. Mira, Sana and Omar occupy other desks, so they are error traps from misplacing the adjacency clue.

Q72. [MBA2-DILR-1 · mcsc · diff 1]

The employee with 10 years of experience works in which department?

- A) HR ✓
- B) Finance
- C) Sales
- D) Legal

Answer: A) HR

Solution: Sales sits at desk 1 with 6 years, so Sana has $6+4=10$ years and is not in Sales. Since Legal holds 8 and Finance is Omar with 4, and the Tech employee holds the least at 2, the only department left for the 10-year employee is HR at desk 5, which is Sana. Legal and Finance are traps from ignoring the experience gap.

Q73. [MBA2-DILR-1 · mcsc · diff 1]

Suppose Omar and Priya exchange only their departments, keeping their desks and their years of experience unchanged. Which statement would then become true?

- A) The Tech employee would end up with the most experience overall.
- B) The Finance and Legal employees would no longer sit adjacently. ✓
- C) The employee at desk 1 would then work in the Legal department.
- D) Exactly two of the five employees would then share one department.

Answer: B) The Finance and Legal employees would no longer sit adjacently.

Solution: Originally Omar is Finance at desk 3 and Priya is Legal at desk 4, satisfying the adjacency rule. After the swap Priya becomes Finance at desk 4 and Omar becomes Legal at desk 3, so Finance now sits to the right of Legal and the two are no longer adjacent in the required order. Desk 1 stays Sales, Tech keeps its 2 years, and each department is still held by exactly one person, so those statements remain false.

Q74. [MBA2-DILR-1 · mcsc · diff 1]

What is the combined experience, in years, of the Finance employee and the Legal employee?

- A) 12 ✓
- B) 16
- C) 10
- D) 14

Answer: A) 12

Solution: Omar is the Finance employee with 4 years, and Priya is the Legal employee with 8 years, so together they hold $4+8=12$ years. The value 14 comes from pairing Legal with Sales, 10 from using Tech instead of Legal, and 16 from adding the two most experienced staff rather than the two named departments.

Instructions (MBA2-DILR-2): study the shared data below and answer its 3 question(s).

Four flights, operated by the airlines Aster, Birch, Cedar and Delta, depart one after another from a regional airport, no two at the same time. Each flight goes to a different city among Lisbon, Cairo, Oslo and Tokyo.

- Aster departs earlier than Birch. - The flight to Tokyo departs first. - Cedar flies to Cairo. - The flight to Oslo departs immediately after the flight to Lisbon. - Delta departs third. - Cedar departs earlier than Delta.

Q75. [MBA2-DILR-2 · mcsc · diff 0.5]

Which airline operates the last flight to depart?

- A) Aster
- B) Delta
- C) Cedar
- D) Birch ✓

Answer: D) Birch

Solution: Tokyo departs first and Aster is earlier than Birch, so Aster takes the first slot with Tokyo. Cedar flies to Cairo and departs before Delta, who is third, so Cedar is second. That leaves Birch for the fourth and final slot. Delta is third and Aster is first, so both are traps for readers who stop after placing Delta.

Q76. [MBA2-DILR-2 · mcsc · diff 0.5]

To which city does Delta fly?

- A) Lisbon ✓
- B) Cairo
- C) Tokyo
- D) Oslo

Answer: A) Lisbon

Solution: Aster (first) goes to Tokyo and Cedar (second) goes to Cairo. Oslo must follow Lisbon immediately, and with Delta third and Birch fourth the only consecutive pair left is Delta then Birch, so Delta flies to Lisbon and Birch to Oslo. Cairo and Tokyo are already assigned, so they are traps for readers who ignore the Lisbon-then-Oslo order.

Q77. [MBA2-DILR-2 · mcsc · diff 1]

Suppose the airport swaps the departure slots of the two flights currently in the second and fourth positions, leaving every destination with its own aircraft unchanged. After the swap, the flight to which city departs immediately before the flight to Oslo?

- A) Cairo
- B) Oslo
- C) Tokyo ✓
- D) Lisbon

Answer: C) Tokyo

Solution: In the original order the slots are Tokyo, Cairo, Lisbon, Oslo. Swapping the second and fourth flights gives Tokyo, Oslo, Lisbon, Cairo. Oslo now sits in the second slot, so the flight departing immediately before it is Tokyo in the first slot. Lisbon and Cairo have moved away from that position, and Oslo cannot precede itself, so they are traps.

Instructions (MBA2-DILR-3): study the shared data below and answer its 4 question(s).

A tour operator sells three package types across four cities. Table 1 gives the number of packages sold (in thousands) and Table 2 gives the price per package (in thousand rupees). Revenue for any cell = packages sold (in '000) x price per package (in '000 rupees), so all revenue figures are in millions of rupees.

****Packages sold (in thousands)****

Package	Jaipur	Kochi	Shimla	Goa
Heritage	4	10	8	6
Coastal	30	8	5	4
Mountain	2	6	10	20

****Price per package (in thousand rupees)****

Package	Jaipur	Kochi	Shimla	Goa
Heritage	20	22	25	18
Coastal	15	16	14	12
Mountain	30	28	26	24

Q78. [MBA2-DILR-3 · mcsc · diff 1]

What is the total revenue earned from Coastal packages across all four cities?

- A) 664 million rupees
- B) 720 million rupees
- C) 696 million rupees ✓
- D) 648 million rupees

Answer: C) 696 million rupees

Solution: Coastal revenue per city is packages sold times price: Jaipur $30 \times 15 = 450$, Kochi $8 \times 16 = 128$, Shimla $5 \times 14 = 70$, Goa $4 \times 12 = 48$, giving 696 million rupees in total. Dropping Goa by mistake leaves 648, which is the tempting near miss.

Q79. [MBA2-DILR-3 · mcsc · diff 1]

The ratio of total Mountain revenue to total Heritage revenue (across all cities) is closest to which of the following?

- A) 7 : 4
- B) 3 : 2
- C) 8 : 5 ✓
- D) 5 : 3

Answer: C) 8 : 5

Solution: Mountain revenue is $60+168+260+480=968$ million rupees and Heritage revenue is $80+220+200+108=608$ million rupees. The ratio 968:608 works out to about 1.59, which is nearest to 8:5 (1.60); 3:2 undershoots at 1.50.

Q80. [MBA2-DILR-3 · mcsc · diff 1]

Which city has the lowest overall average price per package (total revenue divided by total packages sold)?

- A) Goa
- B) Shimla
- C) Kochi
- D) Jaipur ✓

Answer: D) Jaipur

Solution: Overall average price is total revenue over total packages. Jaipur is $590/36=16.4$, Goa $636/30=21.2$, Kochi $516/24=21.5$, Shimla $530/23=23.0$ (thousand rupees). Jaipur is lowest because its huge cheap Coastal volume pulls the weighted price down; taking a plain average of the three listed prices wrongly flags Goa (mean 18).

Q81. [MBA2-DILR-3 · mcsc · diff 1]

By how much does Goa's total revenue exceed Kochi's total revenue?

- A) 96 million rupees
- B) 120 million rupees ✓
- C) 132 million rupees
- D) 108 million rupees

Answer: B) 120 million rupees

Solution: Goa's total revenue is $108+48+480=636$ million rupees and Kochi's is $220+128+168=516$ million rupees, so Goa leads by $636-516=120$ million rupees. The distractors come from small slips in the Mountain or Coastal products.

Instructions (MBA2-DILR-4): study the shared data below and answer its 4 question(s).

A cultural trust runs four community centres, named Aster, Birch, Cedar and Dune. Over one season the four centres together host a total of exactly 26 workshops. Each centre hosts a different whole number of workshops, and each hosts between 3 and 10 workshops (both inclusive). Separately, the four centres open for the season in four different months, one each in March, April, May and June.

The following facts are known about the workshop counts:

- Aster hosts more workshops than every other centre, and Dune hosts fewer than every other centre. - Birch hosts more workshops than Cedar. - Cedar hosts exactly 6 workshops. - Aster hosts an odd number of workshops. - Aster hosts exactly two more workshops than Birch.

The following facts are known about the opening months:

- Cedar opens in the earliest of the four months, and Birch opens in the latest of the four months. - Aster opens in an earlier month than Dune.

Q82. [MBA2-DILR-4 · mcsc · diff 0.5]

How many workshops does Dune host?

- A) 4 ✓
- B) 9
- C) 7
- D) 6

Answer: A) 4

Solution: The distinct counts sum to 26 with Cedar fixed at 6. Aster is the largest and odd, Dune the smallest, Birch above Cedar, and Aster is Birch plus 2. The only fit is Aster 9, Birch 7, Cedar 6, Dune 4, so Dune hosts 4. The 6 trap copies Cedar, 7 copies Birch, and 9 copies Aster.

Q83. [MBA2-DILR-4 · mcsc · diff 1]

Which centre opens in the third of the four months?

- A) Cedar
- B) Birch
- C) Aster
- D) Dune ✓

Answer: D) Dune

Solution: Cedar opens earliest and Birch latest, so those take the first and fourth slots. Aster opens before Dune, which forces Aster into the second slot and Dune into the third. So Dune opens third. Aster is the second-month trap, Cedar is fixed first, and Birch is fixed last.

Q84. [MBA2-DILR-4 · mcsc · diff 1]

Suppose only these workshop facts are kept: the counts are distinct whole numbers from 3 to 10 summing to 26, Aster hosts the most, Dune the fewest, and Birch hosts more than Cedar. The facts fixing Cedar at 6, making Aster odd and setting Aster two above Birch are dropped. How many different count-assignments to the four centres are then possible?

- A) 10
- B) 6
- C) 8 ✓
- D) 12

Answer: C) 8

Solution: With Aster the strict maximum, Dune the strict minimum, Birch above Cedar, distinct values 3 to 10, and a total of 26, a direct enumeration gives exactly 8 valid assignments. The 6 and 10 traps come from stopping the count early or double-counting a mirrored case, and 12 overcounts the middle pair.

Q85. [MBA2-DILR-4 · mcsc · diff 1]

What is the combined number of workshops hosted by the centre that opens first and the centre that opens last?

- A) 16
- B) 11
- C) 15
- D) 13 ✓

Answer: D) 13

Solution: Cedar opens first with 6 workshops and Birch opens last with 7, so the combined total is 13. Using Aster instead of Cedar for the first slot gives the 15 trap, pairing Dune with Birch gives 11, and pairing Aster with Birch gives 16.

Instructions (MBA2-DILR-5): study the shared data below and answer its 3 question(s).

A field consultant visits five clients, named Vernon, Wesley, Xavier, Yara and Zane, over a single working week. She visits exactly one client on each of the five days from Monday to Friday, and no client is visited more than once.

The following conditions all hold:

- Wesley is visited on Monday. - Zane is visited on the day immediately after the day on which Xavier is visited. - Xavier is visited on an earlier day than Yara. - Vernon is visited on an earlier day than Xavier.

Q86. [MBA2-DILR-5 · mcsc · diff 0.5]

Which client does the consultant visit on Friday?

- A) Yara ✓
- B) Zane
- C) Vernon
- D) Xavier

Answer: A) Yara

Solution: Wesley takes Monday. Vernon comes before Xavier, and Zane sits immediately after Xavier, so Vernon, Xavier and Zane fill Tuesday, Wednesday and Thursday in that order. That leaves Yara for Friday, matching the requirement that Xavier precedes Yara. Zane and Xavier fall on Thursday and Wednesday, and Vernon on Tuesday.

Q87. [MBA2-DILR-5 · mcsc · diff 0.5]

On which day is Xavier visited?

- A) Tuesday
- B) Wednesday ✓
- C) Monday
- D) Thursday

Answer: B) Wednesday

Solution: With Wesley on Monday and Vernon before Xavier, Vernon must take Tuesday. Xavier then takes Wednesday, Zane the immediately following Thursday, and Yara Friday. So Xavier is visited on Wednesday. Tuesday belongs to Vernon, Thursday to Zane, and Monday to Wesley.

Q88. [MBA2-DILR-5 · mcsc · diff 1]

Suppose the condition that Vernon is visited before Xavier is dropped, while the other three conditions still hold. How many different visiting orders for the week are then possible?

- A) 4
- B) 5
- C) 3 ✓
- D) 2

Answer: C) 3

Solution: Keeping Wesley on Monday, Zane immediately after Xavier, and Xavier before Yara, the Xavier-Zane block and the remaining two clients can be placed in exactly three valid ways. The 2 trap forgets one placement of Vernon, while 4 and 5 over-count arrangements that break the immediate-after or the Xavier-before-Yara rule.

Instructions (MBA2-DILR-6): study the shared data below and answer its 4 question(s).

A ceramic tile maker ships four tile designs. The first chart shows boxes shipped (in thousands of boxes) and the second shows the price per box (in rupees) for three financial years FY22, FY23 and FY24. Revenue for any design in a year = boxes shipped (in '000) x price per box (in rupees), so all revenue figures are in thousands of rupees.

Q89. [MBA2-DILR-6 · mcsc · diff 1]

What is the total revenue from all four designs in FY24?

- A) 44,300 thousand rupees
- B) 46,200 thousand rupees
- C) 45,710 thousand rupees ✓
- D) 43,950 thousand rupees

Answer: C) 45,710 thousand rupees

Solution: FY24 revenue per design is boxes ('000) times price: Slate $60 \times 220 = 13200$, Marble $45 \times 350 = 15750$, Terra $24 \times 240 = 5760$, Onyx $20 \times 550 = 11000$, summing to 45,710 thousand rupees. The distractors drop or mis-multiply one design.

Q90. [MBA2-DILR-6 · mcsc · diff 1]

Which design recorded the highest percentage growth in revenue from FY22 to FY24?

- A) Terra
- B) Onyx ✓
- C) Marble
- D) Slate

Answer: B) Onyx

Solution: FY22 revenue is Slate 8000, Marble 9000, Terra 5000, Onyx 5000; FY24 is Slate 13200, Marble 15750, Terra 5760, Onyx 11000 (thousand rupees). Growth is Onyx +120%, Marble +75%, Slate +65%, Terra +15%, so Onyx leads. Judging by box counts alone still favours Onyx but hides the size of the gap.

Q91. [MBA2-DILR-6 · mcsc · diff 1]

Over the three years, what is Slate's average price per box weighted by the number of boxes shipped?

- A) 215.00 rupees
- B) 210.00 rupees
- C) 211.33 rupees ✓
- D) 213.33 rupees

Answer: C) 211.33 rupees

Solution: Weighting Slate prices by boxes shipped gives $(40 \times 200 + 50 \times 210 + 60 \times 220) / (40 + 50 + 60) = 31700 / 150 = 211.33$ rupees. A plain average of 200, 210 and 220 ignores volumes and wrongly yields 210.

Q92. [MBA2-DILR-6 · mcsc · diff 1]

By how much did the combined total revenue of all four designs change from FY22 to FY23?

- A) increased by 9,780 thousand rupees
- B) increased by 8,900 thousand rupees
- C) increased by 8,320 thousand rupees
- D) increased by 9,320 thousand rupees ✓

Answer: D) increased by 9,320 thousand rupees

Solution: FY22 combined revenue is $8000+9000+5000+5000=27000$; FY23 is $10500+11520+6500+7800=36320$ (thousand rupees). The rise is $36320-27000=9320$ thousand rupees. The distractors mishandle one design's price or volume.

Section: QA (40 min)

Q93. [MBA1-QA-MCQ-1 · mcsc · diff 0.5]

A dealer marks the price of a ceiling fan 60% above its cost price and then allows two successive discounts of 20% and 10% on the marked price. What is his profit percentage on this sale?

- A) 15.2% ✓
- B) 12.0%
- C) 16.0%
- D) 28.0%

Answer: A) 15.2%

Solution: Take cost as 100, so marked price is 160. After 20% off the price is 128, and after a further 10% off it is 115.2, giving a profit of 15.2 on 100, that is 15.2%. The 12.0% figure comes from treating the two discounts as one flat 30% cut, since $160 \times 0.70 = 112$. The 28.0% figure comes from applying only the 20% discount and stopping. The 16.0% figure comes from reading 10% of 160 as the whole profit.

Q94. [MBA1-QA-MCQ-2 · mcsc · diff 0.5]

Pranav invests Rs 42,000 for 8 months, Qadir invests Rs 56,000 for 5 months and Rohit invests Rs 35,000 for 12 months in a partnership firm. At the end of the year the firm earns a profit of Rs 74,000, shared in proportion to capital multiplied by time. What is Qadir's share?

- A) Rs 24,000
- B) Rs 22,000
- C) Rs 20,000 ✓
- D) Rs 30,000

Answer: C) Rs 20,000

Solution: The capital-months are $42000 \times 8 = 336000$, $56000 \times 5 = 280000$ and $35000 \times 12 = 420000$, which reduce to 12 : 10 : 15, a total of 37 parts. One part is $74000 / 37 = 2000$, so Qadir's 10 parts give Rs 20,000. The Rs 24,000 value is Pranav's 12 parts and the Rs 30,000 value is Rohit's 15 parts, both read off the wrong partner. The Rs 22,000 value comes from mis-simplifying 280 into 11 parts instead of 10.

Q95. [MBA1-QA-MCQ-3 · mcsc · diff 0.5]

A vessel contains 60 litres of a mixture of milk and water in which milk and water are in the ratio 7 : 3. How many litres of water must be added so that the ratio of milk to water in the new mixture becomes 7 : 5?

- A) 15 litres
- B) 12 litres ✓
- C) 8 litres
- D) 10 litres

Answer: B) 12 litres

Solution: The vessel holds 42 litres of milk and 18 litres of water. Milk does not change, so for the ratio 7 : 5 the water must become $42 \times 5 / 7 = 30$ litres, needing $30 - 18 = 12$ litres more. The 10 litres value comes from wrongly making total volume 70. The 15 litres value comes from taking the new water as one quarter of a 60 litre base. The 8 litres value comes from using the ratio difference $5 - 3 = 2$ on 4 litre parts.

Q96. [MBA1-QA-MCQ-4 · mcsc · diff 0.5]

A motorboat whose speed in still water is 12 km/h goes 45 km downstream along a river and returns to the starting point, taking 8 hours in all. What is the speed of the current?

- A) 3 km/h ✓
- B) 2.5 km/h
- C) 4 km/h
- D) 6 km/h

Answer: A) 3 km/h

Solution: If the current is s , then $45 / (12 + s) + 45 / (12 - s) = 8$, so $45 \times 24 = 8 \times (144 - s^2)$, giving $144 - s^2 = 135$ and $s = 3$ km/h. The 4 km/h value comes from averaging the two legs at $45 / 4$ hours each. The 2.5 km/h value comes from solving with a total of 7.5 hours. The 6 km/h value comes from equating the round trip to a single 90 km run at $12 - s$.

Q97. [MBA1-QA-MCQ-5 · mcsc · diff 0.5]

Anil alone can paint a hall in 20 days and Bimal alone can paint it in 30 days. They begin painting together, but Anil stops working 5 days before the hall is finished, and Bimal completes the remaining part alone. In how many days is the hall painted?

- A) 14 days
- B) 12 days
- C) 18 days
- D) 15 days ✓

Answer: D) 15 days

Solution: If the job takes T days, Bimal works all T days and Anil works $T - 5$ days, so $(T - 5) / 20 + T / 30 = 1$. Multiplying by 60 gives $3T - 15 + 2T = 60$, so $T = 15$ days. The 12 days value is the time if both worked throughout. The 14 days value comes from letting Bimal, not Anil, be the one who leaves early. The 18 days value comes from adding 5 idle days to a 13 day estimate.

Q98. [MBA1-QA-MCQ-6 · mcsc · diff 0.5]

A sum of Rs 16,000 is lent for one year at 10% per annum. By how much does the interest earned when the rate is compounded half-yearly exceed the interest earned when it is compounded annually?

- A) Rs 20
- B) Rs 80
- C) Rs 1,640
- D) Rs 40 ✓

Answer: D) Rs 40

Solution: Half-yearly, the amount is $16000 \times 1.05 \times 1.05 = 17640$, so interest is 1640, while annual compounding for one year gives just 1600. The excess is Rs 40. The Rs 80 value comes from using 10% per half year instead of 5%. The Rs 20 value comes from halving the correct gap. The Rs 1,640 value is the half-yearly interest itself, not the excess over annual compounding.

Q99. [MBA1-QA-MCQ-7 · mcsc · diff 0.5]

Four boys and three girls are to be seated in a row of seven chairs. In how many ways can they be seated if no two girls sit next to each other?

- A) 5040
- B) 720
- C) 2880
- D) 1440 ✓

Answer: D) 1440

Solution: Seat the four boys first in $4! = 24$ ways, creating five gaps including the two ends. Choose and fill three of those gaps with the three girls in $5 \times 4 \times 3 = 60$ ordered ways, giving $24 \times 60 = 1440$. The 720 value drops the ordering of the girls by dividing by $3!$. The 2880 value comes from allowing six gaps instead of five. The 5040 value is $7!$, the count with no restriction at all.

Q100. [MBA1-QA-MCQ-8 · mcsc · diff 1]

Bag A contains 4 red and 6 green balls, while bag B contains 7 red and 3 green balls. One of the two bags is chosen at random and two balls are drawn from it together. Both drawn balls turn out to be red. What is the probability that bag A was the bag chosen?

- A) $\frac{1}{3}$
- B) $\frac{2}{9}$ ✓
- C) $\frac{7}{9}$
- D) $\frac{2}{15}$

Answer: B) $\frac{2}{9}$

Solution: Each bag has 10 balls, so two reds have chance $\frac{6}{45}$ from bag A and $\frac{21}{45}$ from bag B. Since the bags are equally likely, the required probability is $\frac{6}{6 + 21} = \frac{2}{9}$. The $\frac{2}{15}$ value is the unconditional chance of two reds from bag A alone. The $\frac{7}{9}$ value is the probability for bag B, the complement. The $\frac{1}{3}$ value comes from using $\frac{4}{10}$ and $\frac{7}{10}$ as single-draw odds and rounding the ratio.

Q101. [MBA1-QA-MCQ-9 · mcsc · diff 1]

A solid metal cone of base radius 6 cm and height 24 cm is melted down, and 10% of the metal is lost in the process. The remaining metal is recast into solid spheres of radius 2 cm each. What is the largest number of complete spheres that can be made?

- A) 8
- B) 25
- C) 24 ✓
- D) 27

Answer: C) 24

Solution: The cone has volume 288π cubic cm, and 90% of it is 259.2π . Each sphere takes $\frac{32\pi}{3}$ cubic cm, so $259.2 \times \frac{3}{32} = 24.3$, and only 24 complete spheres are possible. The 27 value ignores the 10% metal loss. The 25 value comes from rounding 24.3 upward instead of down. The 8 value comes from omitting the factor $\frac{1}{3}$ in the volume of a sphere.

Q102. [MBA1-QA-MCQ-10 · mcsc · diff 1]

Let N be the number whose prime factorisation is $2^a \times 3^b \times 5^c$. How many positive divisors of N are perfect squares and are also multiples of 12?

- A) 12
- B) 8 ✓
- C) 4
- D) 6

Answer: B) 8

Solution: A divisor $2^a \times 3^b \times 5^c$ is a perfect square only when a, b and c are all even, and it is a multiple of 12 only when a is at least 2 and b is at least 1. So a is 2 or 4, b is 2 or 4, and c is 0 or 2, giving $2 \times 2 \times 2 = 8$ divisors. The 12 value allows b = 0 as well, dropping the multiple of 3 condition. The 6 value allows b = 0 but also forces c = 0. The 4 value forgets that c can be 2.

Q103. [MBA1-QA-MCQ-11 · mcsc · diff 1]

Three positive numbers are in arithmetic progression. Their sum is 27 and the sum of their squares is 293. What is the product of the three numbers?

- A) 405
- B) 585
- C) 504 ✓
- D) 729

Answer: C) 504

Solution: Write the numbers as $9 - d$, 9 and $9 + d$, since the sum 27 forces the middle term to be 9. Then $243 + 2d^2 = 293$, so $d^2 = 25$ and the numbers are 4, 9 and 14, whose product is 504. The 729 value comes from taking all three numbers equal to 9, which fits the sum but not the square sum. The 585 value uses $d = 4$ and the 405 value uses $d = 6$, neither of which gives 293.

Q104. [MBA1-QA-MCQ-12 · mcsc · diff 1]

A non-zero real number x satisfies $x + 1/x = 3$. What is the value of $x^5 + 1/x^5$?

- A) 141
- B) 243
- C) 123 ✓
- D) 126

Answer: C) 123

Solution: Write $s(n) = x^n + 1/x^n$. Multiplying $s(n)$ by $(x + 1/x)$ gives $s(n+1) + s(n-1)$, so $s(n+1) = 3 s(n) - s(n-1)$ with $s(0) = 2$ and $s(1) = 3$. This gives $s(2) = 7$, $s(3) = 18$, $s(4) = 47$ and $s(5) = 123$. The value 243 comes from assuming the answer is 3^5 , that is from treating $x + 1/x$ as if it were x itself. The value 126 is the product 7 times 18, an invented shortcut $s(2)$ times $s(3)$. The value 141 follows from the slip $s(4) = 3$ times $18 - 3 = 51$ and then 3 times $51 - 18$.

Q105. [MBA1-QA-MCQ-13 · mcsc · diff 1]

If a and b are positive real numbers with $a + b = 10$, what is the maximum possible value of $a^2 b^3$?

- A) 3087
- B) 4000
- C) 3125
- D) 3456 ✓

Answer: D) 3456

Solution: Split the sum as $a/2 + a/2 + b/3 + b/3 + b/3$, five positive terms adding to 10. By AM-GM the product $(a^2 b^3)/108$ is largest when all five terms are equal, so $a/2 = b/3 = 2$, giving $a = 4$ and $b = 6$. Then $a^2 b^3 = 16$ times $216 = 3456$. The value 3125 comes from setting $a = b = 5$, which maximises only symmetric products. The value 4000 comes from the guess $a = 2$, $b = 8$ followed by 4 times 1000. The value 3087 corresponds to $a = 3$ and $b = 7$, a nearby but smaller choice.

Q106. [MBA1-QA-MCQ-14 · mcsc · diff 1]

The equation $(\log_3 x)^2 + 3 = \log_3 (x^4)$ holds for exactly two positive values of x . What is the sum of these two values?

- A) 30 ✓
- B) 84
- C) 12
- D) 28

Answer: A) 30

Solution: Put $t = \log_3 x$. Since $\log_3 (x^4) = 4t$, the equation becomes $t^2 - 4t + 3 = 0$, so $t = 1$ or $t = 3$, giving $x = 3$ and $x = 27$, whose sum is 30. The value 84 comes from misreading the roots as $t = 1$ and $t = 4$, which gives $3 + 81$. The value 12 comes from adding the roots 1 and 3 after cubing only one of them, that is $9 + 3$. The value 28 comes from dropping the constant 3 and using $t = 0$ and $t = 3$, so $1 + 27$.

Q107. [MBA1-QA-MCQ-15 · mcsc · diff 1]

In the coordinate plane, $A = (0, 2)$ and $B = (12, 3)$. A point P moves along the x -axis. What is the smallest possible value of $PA + PB$?

- A) 13 ✓
- B) 17
- C) 15
- D) 12

Answer: A) 13

Solution: Reflect A in the x -axis to get $A' = (0, -2)$. For any P on the axis $PA = PA'$, so $PA + PB$ equals $PA' + PB$, which is least when P lies on the segment $A'B$. The minimum is the length $A'B = \sqrt{12^2 + 5^2} = 13$, reached at $P = (4.8, 0)$. The value 17 comes from adding the two heights and the horizontal gap, $2 + 3 + 12$. The value 12 is only the horizontal gap between A and B . The value 15 comes from subtracting the heights instead of adding them and then rounding $\sqrt{12^2 + 1^2}$ upward.

Q108. [MBA1-QA-MCQ-16 · mcsc · diff 1]

Among 200 students, 120 play cricket, 90 play football and 70 play tennis. Exactly 40 play both cricket and football, 30 play both football and tennis, 35 play both cricket and tennis, and 15 play all three games. How many students play exactly one of the three games?

- A) 145
- B) 115 ✓
- C) 130
- D) 100

Answer: B) 115

Solution: By inclusion and exclusion the number playing at least one game is $120 + 90 + 70 - 40 - 30 - 35 + 15 = 190$. The pairwise counts include the 15 triple players, so the numbers playing exactly two games are $40 - 15 = 25$, $30 - 15 = 15$ and $35 - 15 = 20$, a total of 60. Subtracting the 60 and the 15 from 190 leaves 115 who play exactly one game. The value 130 comes from subtracting only 60 and forgetting the triple group. The value 100 comes from subtracting 2 times 15 twice over. The value 145 treats the pairwise figures as exactly-two counts already.

Q109. [MBA1-QA-MCQ-17 · mcsc · diff 1]

What is the value of the sum $2/(1 \cdot 3) + 2/(3 \cdot 5) + 2/(5 \cdot 7) + \dots + 2/(47 \cdot 49)$?

- A) 24/25
- B) 48/49 ✓
- C) 47/49
- D) 24/49

Answer: B) 48/49

Solution: Each term splits as $2/((2k-1)(2k+1)) = 1/(2k-1) - 1/(2k+1)$, so consecutive parts cancel. With the last denominator pair being 47 and 49, k runs from 1 to 24 and the sum collapses to $1 - 1/49 = 48/49$. The value 24/49 comes from halving the telescoped result, that is from keeping the factor 1/2 that the numerator 2 has already cancelled. The value 47/49 comes from stopping the cancellation one term early at $1 - 2/49$. The value 24/25 comes from miscounting the terms and telescoping to $1 - 1/25$.

Q110. [MBA1-QA-TITA-1 · tita · diff 0.5]

What is the sum of all the odd positive factors of 1800? Type your answer as an integer.

Answer (TITA): 403

Solution: 1800 factorises as $2^3 \times 3^2 \times 5^2$. An odd factor uses no power of 2, so the odd factors are exactly the factors of $3^2 \times 5^2 = 225$. Their sum is $(1 + 3 + 9)(1 + 5 + 25) = 13 \times 31 = 403$. Using the full factor sum $(1 + 2 + 4 + 8)(13)(31) = 6045$ wrongly keeps the even factors, and 13×31 miscounted as $12 \times 30 = 360$ comes from dropping the term 1 in each bracket.

Q111. [MBA1-QA-TITA-2 · tita · diff 0.5]

In a colony of 120 households, every household subscribes to at least one of three newspapers A, B and C. Exactly 70 subscribe to A, 60 subscribe to B and 50 subscribe to C, while 12 households subscribe to all three. How many households subscribe to exactly two of the three newspapers? Type your answer as an integer.

Answer (TITA): 36

Solution: By inclusion and exclusion, $120 = 70 + 60 + 50$ minus (sum of the three pairwise overlaps) plus 12, so the pairwise overlap total is $180 + 12 - 120 = 72$. Each pairwise overlap counts the 12 all-three households once, so the exactly-two count is $72 - 3 \times 12 = 36$. Answering 72 stops at the raw pairwise total, and 48 comes from subtracting 12 only once instead of three times.

Q112. [MBA1-QA-TITA-3 · tita · diff 1]

What is the remainder when $3^{2024} + 4^{2024}$ is divided by 100? Type your answer as an integer.

Answer (TITA): 37

Solution: The last two digits of powers of 3 repeat with period 20, and 2024 leaves remainder 4 on division by 20, so 3^{2024} ends in the same two digits as $3^4 = 81$. The last two digits of powers of 4 repeat with period 10 from the first power onward, and 2024 leaves remainder 4 on division by 10, so 4^{2024} ends like $4^4 = 256$, that is 56. Then $81 + 56 = 137$, whose remainder on division by 100 is 37. Answering 137 forgets the final reduction, and 97 comes from using a period of 4 for powers of 4 and getting 16.

Q113. [MBA1-QA-TITA-4 · tita · diff 1]

The positive integers x and y satisfy $x + y = 30$. What is the largest possible value of $x^2 y$? Type your answer as an integer.

Answer (TITA): 4000

Solution: Write the product as $(x/2)(x/2)y$, three quantities whose sum is $x + y = 30$ and hence fixed. The product of such quantities is largest when they are equal, giving $x/2 = y$, so $x = 20$ and $y = 10$, and the value is $400 \times 10 = 4000$. Checking the integer neighbours confirms it, since $19^2 \times 11 = 3971$ and $21^2 \times 9 = 3969$. Answering 3375 comes from wrongly setting $x = y = 15$, and 8100 comes from maximising x^2 alone at $x = 27$.

Q114. [MBA1-QA-TITA-5 · tita · diff 1]

In how many distinct arrangements of all eleven letters of the word MISSISSIPPI do no two letters S stand next to each other? Type your answer as an integer.

Answer (TITA): 7350

Solution: Set the four letters S aside and arrange the remaining seven letters M, I, I, I, I, P, P, which can be done in $7!/(4! \times 2!) = 105$ ways. Those seven letters create eight gaps, and choosing four of them for the four identical letters S keeps every pair of S separated, giving $C(8,4) = 70$ choices. The count is $105 \times 70 = 7350$. Answering 34650 is the total number of arrangements with no separation condition, and 8400 comes from using $C(8,4)$ with 120 instead of 105 for the non-S block.

Q115. [MBA2-QA-MCQ-1 · mcsc · diff 0.5]

A trader marks his goods 60% above cost price and then allows two successive discounts of 25% and 20% on the marked price. What is his overall profit or loss percentage?

- A) 4% loss ✓
- B) 6% loss
- C) 4% profit
- D) 2% loss

Answer: A) 4% loss

Solution: Take cost 100, so marked price is 160. After 25% off it is 120, and after a further 20% off it is 96. Selling at 96 against cost 100 is a 4% loss. The 4% profit trap forgets one discount, and the 6% figure comes from misapplying the second discount to cost.

Q116. [MBA2-QA-MCQ-2 · mcsc · diff 0.5]

A invests Rs 16000 for 9 months, B invests Rs 24000 for 4 months and C invests Rs 12000 for 10 months in a business. If the total annual profit is Rs 9000, what is B's share?

- A) Rs 3600
- B) Rs 3000
- C) Rs 2400 ✓
- D) Rs 2000

Answer: C) Rs 2400

Solution: Capital-months are 144000, 96000 and 120000, which reduce to the ratio 6:4:5 summing to 15 parts. B holds 4 parts, so B earns 9000 times 4 over 15, equal to Rs 2400. The Rs 3000 trap splits equally, and Rs 3600 uses the wrong part count.

Q117. [MBA2-QA-MCQ-3 · mcsc · diff 0.5]

A vessel holds 60 litres of a milk and water mixture in which milk and water are in the ratio 5:1. How much water must be added so that the ratio of milk to water becomes 5:2?

- A) 12 litres
- B) 10 litres ✓
- C) 15 litres
- D) 8 litres

Answer: B) 10 litres

Solution: The 60 litres contain 50 litres milk and 10 litres water. Milk stays fixed at 50, and for a 5:2 ratio the water needed is 50 times 2 over 5, which is 20 litres. Since 10 litres are already present, 10 more must be added. The 20 figure forgets the existing water.

Q118. [MBA2-QA-MCQ-4 · mcsc · diff 0.5]

A boat covers 36 km downstream in 2 hours and the same 36 km upstream in 3 hours. What is the speed of the stream?

- A) 6 km/h
- B) 3 km/h ✓
- C) 2 km/h
- D) 15 km/h

Answer: B) 3 km/h

Solution: Downstream speed is 36 over 2, that is 18 km/h, and upstream speed is 36 over 3, that is 12 km/h. Stream speed is half the difference, so it equals half of 6, which is 3 km/h. The 15 value wrongly reports boat speed, and 6 forgets to halve the difference.

Q119. [MBA2-QA-MCQ-5 · mcsc · diff 0.5]

A can finish a piece of work in 8 days and B can finish the same work in 12 days. They work on alternate days with A starting on the first day. In how many days will the work be completed?

- A) 9.5 days ✓
- B) 9 days
- C) 10 days
- D) 8.5 days

Answer: A) 9.5 days

Solution: With 24 total units, A does 3 units per day and B does 2 units per day. Over 9 alternating days starting with A the completed work is 23 units. One unit remains for B, who needs half a day at 2 units per day, giving 9.5 days in all. The 10 trap rounds up the final partial day.

Q120. [MBA2-QA-MCQ-6 · mcsc · diff 0.5]

A sum of Rs 8000 is invested at 20% per annum compounded half-yearly for one year. What is the compound interest earned?

- A) Rs 1600
- B) Rs 1728
- C) Rs 1764
- D) Rs 1680 ✓

Answer: D) Rs 1680

Solution: Half-yearly the rate is 10% over two periods, so the amount is 8000 times 1.1 squared, that is 9680. The interest is 9680 minus 8000, which is Rs 1680. The Rs 1600 value uses simple interest, and Rs 1764 wrongly applies a 20% rate for two full periods.

Q121. [MBA2-QA-MCQ-7 · mcsc · diff 0.5]

In how many ways can the letters of the word PENCIL be arranged so that the two vowels E and I are never together?

- A) 480 ✓
- B) 720
- C) 240
- D) 600

Answer: A) 480

Solution: All six distinct letters arrange in 720 ways. Treating the two vowels as one block gives 5 factorial times 2, that is 240 arrangements where they sit together. Subtracting leaves 720 minus 240, which is 480. The 240 value counts the together case instead of the apart case.

Q122. [MBA2-QA-MCQ-8 · mcsc · diff 1]

A bag contains 5 red balls and 4 blue balls. Two balls are drawn at random without replacement. What is the probability that at least one of them is red?

- A) 5/6 ✓
- B) 5/9
- C) 1/6
- D) 2/3

Answer: A) 5/6

Solution: The complement of at least one red is both balls blue, which has probability 6 over 36, that is 1/6, using choose two from four over choose two from nine. Therefore at least one red is 1 minus 1/6, which is 5/6. The 1/6 value reports the complement, and 5/9 uses a single draw.

Q123. [MBA2-QA-MCQ-9 · mcsc · diff 1]

A solid metallic sphere of radius 6 cm is melted and recast into small cones, each having base radius 2 cm and height 3 cm. How many such cones can be made?

- A) 72 ✓
- B) 36
- C) 54
- D) 108

Answer: A) 72

Solution: The sphere volume uses four-thirds times radius cubed, giving 288 times pi. Each cone volume uses one-third times base area times height, giving 4 times pi. Dividing 288 by 4 yields 72 cones. The 36 trap uses half the sphere volume, and 108 mishandles the cone factor.

Q124. [MBA2-QA-MCQ-10 · mcsc · diff 1]

What is the remainder when 3 raised to the power 101 is divided by 100?

- A) 9
- B) 3 ✓
- C) 1
- D) 27

Answer: B) 3

Solution: Powers of 3 modulo 100 cycle with period 20, since 3 to the power 20 leaves remainder 1. As 101 leaves remainder 1 when divided by 20, the value equals 3 to the power 1 modulo 100, which is 3. The 1 trap forgets the leftover exponent, and 27 uses exponent 3.

Q125. [MBA2-QA-MCQ-11 · mcsc · diff 1]

The sum of the first n terms of an arithmetic progression is given by $3n^2$ plus $5n$. What is the 12th term of this progression?

- A) 68
- B) 74 ✓
- C) 80
- D) 72

Answer: B) 74

Solution: The nth term equals the sum of n terms minus the sum of n minus one terms. Here that difference simplifies to $6n$ plus 2, so the 12th term is 72 plus 2, which is 74. The 72 trap drops the constant, and 80 mishandles the sum-difference step.

Q126. [MBA2-QA-MCQ-12 · mcsc · diff 1]

If $a + b + c = 0$ and $a^2 + b^2 + c^2 = 14$, what is the value of $a^2b^2 + b^2c^2 + c^2a^2$?

- A) 196
- B) 49
- C) 98 ✓
- D) 72

Answer: C) 98

Solution: With $a+b+c=0$, we get $ab+bc+ca = -(a^2+b^2+c^2)/2 = -7$, so $(ab+bc+ca)^2$ gives $a^2b^2+b^2c^2+c^2a^2 = 49$ (the abc *sum term vanishes). Then $a^2b^2+b^2c^2+c^2a^2 = (a^2+b^2+c^2)^2 - 2(a^2b^2+b^2c^2+c^2a^2) = 196 - 98 = 98$. The 196 value forgets to subtract the cross terms, 49 stops at the cross-term sum, and 72 mishandles the squaring.

Q127. [MBA2-QA-MCQ-13 · mcsc · diff 1]

For $x > 0$, what is the minimum value of $x^2 + 250/x$?

- A) 100
- B) 70
- C) 75 ✓
- D) 80

Answer: C) 75

Solution: Split the second term to apply AM-GM on three parts: $x^2 + 125/x + 125/x$ is at least 3 times the cube root of $(x^2 * 125/x * 125/x) = 3 * \text{cube root of } 15625 = 3 * 25 = 75$, with equality when $x^3 = 125$, so $x = 5$. Splitting into only two equal pieces gives the wrong 100 figure.

Q128. [MBA2-QA-MCQ-14 · mcsc · diff 1]

What is the value of the product $(\log \text{ base 2 of } 3)(\log \text{ base 3 of } 4)(\log \text{ base 4 of } 5) \dots (\log \text{ base 63 of } 64)$?

- A) 5
- B) 8
- C) 32
- D) 6 ✓

Answer: D) 6

Solution: Writing each factor as $\ln$ of the top over $\ln$ of the base makes the chain telescope, leaving $\ln 64$ divided by $\ln 2$, which equals $\log \text{ base 2 of } 64 = 6$ since $2^6 = 64$. The 32 trap divides 64 by 2, and 5 miscounts the exponent.

Q129. [MBA2-QA-MCQ-15 · mcsc · diff 1]

Points A(2, 3) and B(8, 5) lie in the coordinate plane. A point P is chosen on the x-axis. What is the least possible value of AP + PB?

- A) 12
- B) $\sqrt{52}$
- C) $\sqrt{68}$
- D) 10 ✓

Answer: D) 10

Solution: Reflect A across the x-axis to A'(2, -3); then AP equals A'P, so AP + PB is least when A', P, B are collinear, giving the straight distance A'B = $\sqrt{(8-2)^2 + (5+3)^2} = \sqrt{36+64} = 10$. Using the direct distance AB gives the $\sqrt{52}$ trap.

Q130. [MBA2-QA-MCQ-16 · mcsc · diff 1]

Among 100 employees, 40 use Python, 48 use R, and 45 use SQL. Also 15 use both Python and R, 18 use both R and SQL, 12 use both Python and SQL, and 8 use all three tools. How many employees use exactly one of the three tools?

- A) 67 ✓
- B) 96
- C) 75
- D) 59

Answer: A) 67

Solution: The number in at least one is $40+48+45 - 15-18-12 + 8 = 96$. Those in exactly two pairs number $(15+18+12) - 3*8 = 21$, and 8 use all three, so exactly one is $96 - 21 - 8 = 67$. The 96 trap stops at the union total, and 59 subtracts the pairwise counts without adding back the triple.

Q131. [MBA2-QA-MCQ-17 · mcsc · diff 1]

What is the sum $\frac{1}{1*3} + \frac{1}{3*5} + \frac{1}{5*7} + \dots + \frac{1}{19*21}$?

- A) $\frac{9}{19}$
- B) $\frac{10}{21}$ ✓
- C) $\frac{20}{21}$
- D) $\frac{11}{23}$

Answer: B) $\frac{10}{21}$

Solution: Each term $\frac{1}{(2k-1)(2k+1)}$ equals one half times $(\frac{1}{2k-1} - \frac{1}{2k+1})$, so the ten terms telescope to one half times $(\frac{1}{1} - \frac{1}{21}) = \text{one half times } \frac{20}{21} = \frac{10}{21}$. Forgetting the factor of one half gives the $\frac{20}{21}$ trap.

Q132. [MBA2-QA-TITA-1 · tita · diff 0.5]

Find the sum of all positive integer divisors of 360 (including 1 and 360 themselves).

Answer (TITA): 1170

Solution: Write 360 as 2^3 times 3^2 times 5. The sum of divisors is multiplicative, so it equals $(1+2+4+8)$ times $(1+3+9)$ times $(1+5)$, that is 15 times 13 times 6, which is 1170. A common slip is to forget the constant term 1 in one bracket, giving 14 times 13 times 6 or 1092, or to stop at 2^2 and use $(1+2+4)$ which yields 7 times 13 times 6 or 546. The full factorisation gives 1170.

Q133. [MBA2-QA-TITA-2 · tita · diff 0.5]

An arithmetic progression has first term 7 and common difference 4. Find the sum of its first 30 terms.

Answer (TITA): 1950

Solution: The sum of the first n terms is $(n/2)$ times $(2a + (n-1)d)$. With a equal to 7, d equal to 4 and n equal to 30, this is 15 times $(14 + 29 \text{ times } 4)$, that is 15 times $(14 + 116)$, which is 15 times 130, giving 1950. Using $(n-1)$ as 30 instead of 29 gives 15 times 134 or 2010, and forgetting to double a gives 15 times $(7 + 116)$ or 1845. The correct sum is 1950.

Q134. [MBA2-QA-TITA-3 · tita · diff 1]

What is the remainder when 17 raised to the power 256 is divided by 100? (In other words, give the last two digits of 17^{256} .)

Answer (TITA): 81

Solution: The last two digits of powers of 17 repeat with a cycle length of 20, since 17^{20} ends in 01 modulo 100. Reducing the exponent, 256 leaves remainder 16 when divided by 20, so 17^{256} has the same last two digits as 17^{16} . Squaring step by step modulo 100 gives 17^2 as 89, 17^4 as 21, 17^8 as 41, and 17^{16} as 81. Treating the cycle length as 4 (as if it were 7) mistakes it, and assuming 256 is a multiple of the period wrongly yields 01. The answer is 81.

Q135. [MBA2-QA-TITA-4 · tita · diff 1]

In a group of 200 people, 100 use app A, 90 use app B and 80 use app C. Exactly 40 people use all three apps and exactly 20 people use none of the three. How many people use exactly two of the three apps?

Answer (TITA): 10

Solution: The number using at least one app is 200 minus 20, that is 180. By inclusion and exclusion, at least one equals the sum of singles minus the sum of the three pairwise overlaps plus the triple overlap, so 180 equals 270 minus (sum of pairwise) plus 40, giving the sum of pairwise overlaps as 130. Each pairwise overlap counts the all-three group as well, so the sum of pairwise overlaps equals (exactly two) plus three times 40. Thus exactly two equals 130 minus 120, which is 10. Forgetting to subtract the triple three times leaves 130, and ignoring the 20 with none gives a different single total. The answer is 10.

Q136. [MBA2-QA-TITA-5 · tita · diff 1]

A positive integer 30 is to be written as a sum of one or more positive integers. Over all such ways, what is the maximum possible value of the product of those parts?

Answer (TITA): 59049

Solution: To maximise the product for a fixed sum, break the number into parts of size 3 as much as possible, avoiding parts of 1, and use a 2 only when the leftover after taking threes is 1 (by combining 3 plus 1 into 2 plus 2). Here 30 is exactly ten 3s with no remainder, so the best product is 3 raised to the power 10, which is 59049. Using all 2s gives 2 raised to the power 15 or 32768, which is smaller, and mixing in a part of 1 only lowers the product. The maximum is 59049.